



Preclinical Characterization and Phase I Study of an Anti-HER2-TLR7 Immune-Stimulator Antibody Conjugate in Patients with HER2⁺ Malignancies

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ABSTRACT

Immune-stimulator antibody conjugates (ISAC) combining tumor-targeting monoclonal antibodies with immunostimulatory agents allow targeted delivery of immune activators into tumors. NJH395 is a novel, first-in-class ISAC comprising a Toll-like receptor 7 (TLR7) agonist conjugated to an anti-HER2 antibody via a noncleavable linker payload. Preclinical characterization showed ISAC-mediated activation of myeloid cells in the presence of antigen-expressing cancer cells, with antigen targeting and TLR7 agonism contributing to antitumor activity. Safety, efficacy, immunogenicity, pharmacokinetics, and pharmacodynamics were inves-

tigated in a phase I, multicenter, open-label study in patients with HER2⁺ non-breast advanced malignancies (NCT03696771). Data from 18 patients enrolled in single ascending dose escalation demonstrated delivery of the TLR7-agonist payload in HER2⁺ tumor cells and induction of type I IFN responses, which correlated with immune modulation in the tumor microenvironment. Cytokine release syndrome was a common, but manageable, drug-related adverse event. Antidrug antibodies and neuroinflammation at high doses represented significant clinical challenges. Data provide proof-of-mechanism and critical insights for novel immunotherapies.

Introduction

Toll-like receptors (TLR) belong to the family of pattern recognition receptors (PRR), a group of proteins that boost immunity by recognizing pathogen-associated molecular patterns (PAMP) and damage-associated molecular patterns (DAMP; refs. 1–3). TLR7 is a type 1 transmembrane glycoprotein in the endosomal membrane of macrophages, monocytes, B lymphocytes, and plasmacytoid dendritic cells (DC; refs. 4–6). Upon ligand binding, TLR7 undergoes conformational changes that result in NF- κ B-mediated expression of inflammatory cytokines, such as interleukin (IL)1, IL6, and type 1 IFNs (7–9). TLR agonists induce cytokine responses that promote polarization of the tumor microenvironment (TME; ref. 10) and antitumor immunity (11–13) by enhancing CD8⁺ T-cell function, inhibiting T-regulatory

cells (Treg; ref. 11), and promoting the maturation of myeloid-derived suppressor cells (MDSC; refs. 14, 15).

Despite reported successes with topical formulations of synthetic TLR7 agonists for certain skin cancers (16–18), systemic administration has demonstrated marginal results, as dosing is limited by toxicities related to systemic immune activation (19–21). Therefore, efforts focus largely on intratumoral administration of TLR agonists. Preliminary results from early-phase trials provide evidence of clinical efficacy, with shrinkage of injected and noninjected lesions (22) and increases in tumoral CD8⁺ T cells (23). Although intratumoral administration of TLR7 agonists is applicable to superficial tumors, it is challenging in patients with deep-seated lesions (23). The potential and challenges of TLR7 targeting highlight the need for novel drugs that enable selective delivery to tumors, allowing adequate concentrations of drug to attract immunostimulatory cells to the TME, while limiting systemic inflammatory toxicities. Immune-stimulator antibody conjugates (ISAC) combining tumor-targeting monoclonal antibodies with potent immunostimulatory agents represent a promising strategy for the targeted delivery of immune activators in tumors. NJH395 is a novel, first-in-class ISAC comprising a TLR7 agonist conjugated to a HER2 antibody through a noncleavable linker payload. This study describes the preclinical characterization and safety, efficacy, immunogenicity, pharmacokinetic (PK), and pharmacodynamic findings from the phase I clinical trial of NJH395 in patients with HER2⁺ non-breast advanced malignancies.

Materials and Methods

Cell lines

The THP-1 cell line for coculture assays (ATCC TIB-202) was expanded in growth media [RPMI (Gibco, cat. #11875-085) supplemented with 10% FBS (Avantor, cat. #97068-085), 1% penicillin–streptomycin (Gibco, cat. #15070-063), and 0.05 mmol/L β -mercaptoethanol

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(Thermo Fisher, cat. #21985023)] to make an assay cell bank. Cells were subcultured every 2 to 4 days, were not allowed to exceed 1×10^6 cells/mL and used at passages 5 to 12 for coculture assay. Cell lines N87 (cat. #CRL-5822; LOT#7686255), HCC1954 (cat. #CRL-2338; LOT#5107643), and SKOV-3 (ATCC cat. #HTB-77; LOT#7349765) were purchased from the ATCC. The HCC1954 and N87 cell lines were expanded in RPMI supplemented with 10% FBS (and 1% penicillin–streptomycin for *in vitro* use). N87 cells for tumor studies were started from a stock tested for *Mycoplasma* and murine viruses in 2010, passaged every 3 to 4 days, and harvested at passage 16. HCC1954 cells for tumor studies were started from a stock tested for *Mycoplasma* and murine viruses in 2014, passaged every 3 to 4 days, and harvested at passage 17. SKOV-3 cells were cultured in McCoy's 5A media supplemented with 10% FBS from a stock tested for *Mycoplasma* and murine viruses in 2015; the stocks were passaged every 3 to 4 days and harvested at passage 11. The HEK293T-TLR7-NFkB-luciferase and HEK293T-TLR7-NFkB-luciferase cell lines were expanded in DMEM High Glucose (Thermo Fisher, HyClone) supplemented with 2 mmol/L glutamine (Thermo Fisher), 10% heat-inactivated FBS, 100 units/mL of penicillin–streptomycin ($1 \times$; Thermo Fisher), and 5 μ g/mL blasticidin (InvivoGen) in T175 flasks to 75% to 90% confluence prior to use. The THP-1-Dual reporter cell line (InvivoGen) was grown in DMEM High Glucose (Thermo Fisher HyClone) supplemented with 10% FBS (Omega Scientific), 2-mmol/L glutamine (Thermo Fisher), 100 U/mL penicillin–streptomycin (Thermo Fisher), 10-mmol/L HEPES (Thermo Fisher), 0.1 mg/mL sodium pyruvate (Thermo Fisher), 0.075% sodium bicarbonate (Thermo Fisher), and 100 mg/mL Normocin (InvivoGen), with 100 mg/mL Zeocin (InvivoGen), and 10 mg/mL blasticidin (InvivoGen) supplemented in every other cell split. The cells were passaged every 2 to 3 days at 7×10^5 cells/mL. Cells were generally not allowed to reach concentrations above 2×10^6 cells/mL, and time spent at room temperature was kept to a minimum. Cells were used at low passage numbers and discarded once they reached passage 20.

Synthesis of NJH395, variants, and MIW338

Antibodies for preclinical pharmacology research were expressed from a two plasmid system encoding the heavy and light antibody chains. Standard site-directed mutagenesis methods were used to introduce amino acid changes to produce variants as described in the text (mutations are described in EU numbering convention). The antibodies were expressed in a CHO cell line and purified using standard affinity chromatography and preparative size exclusion chromatography as needed. Conjugations were performed with standard thiol/maleimide chemistry and ISACs were analyzed to confirm quality appropriate for the intended uses.

Catabolite extraction

N87 cells were cultured in the presence of 2 μ g/mL NJH395 for 48 hours. Media were collected, filtered, and frozen until analysis. Cells were scraped off the plates, washed with PBS, pelleted by centrifugation, and frozen. After thawing on ice, cell aliquots were made and spiked with synthetic compounds to make standard samples for expected catabolites. Samples were then extracted by mixing with 75/25 (v/v) acetonitrile/methanol and incubating for 60 minutes at -20°C . The precipitate was removed by centrifugation for 10 minutes at 15,000 rcf. Media were thawed, concentrated 4-fold, and then treated similarly to cell pellet samples. Standards were dried and reconstituted in a defined volume of 5% acetonitrile and 0.1% formic acid.

Catabolite mass spectrometry analysis

The liquid chromatography–mass spectrometry (LC-MS) analysis of extracted cell pellets was performed using a 6550 iFunnel QTOF LC-MS high-resolution mass spectrometer (Agilent) coupled to a 1260 Infinity UHPLC system (Agilent) containing a binary separation module with a cooled autosampler. For chromatography, an Aeris Widepore XB-C18, 50 mm $\times$ 2.1 mm, 3.6 microparticle size LC column (Phenomenex) was operated at 40°C and 0.4 mL/minute. Runs comprised a linear gradient from 5% to 60% acetonitrile, with 0.1% formic acid with a wash step at 90% acetonitrile and 0.1% formic acid between samples. The MS analysis was done in a positive mode using a Dual Agilent Jet Stream ESI source. The nebulizing gas temperature was set to 200°C , and the drying gas flow rate to 17 L/minute with a nebulizer pressure of 45 psig. The sheath gas temperature was 300°C , and the sheath gas flow rate was 12 L/minute. The V_{Cap} voltage of the source was 2.5 kV, nozzle voltage was 2 kV, MS-TOF fragmentor voltage was 300 V, and the transfer octupole RF peak-to-peak voltage V_{pp} was 750 V. Putative catabolite verification was based on the retention time and mass compared with isotopically labeled internal standards. Quantification was performed by generation of the extracted ion chromatograms (EIC) and integration of the monoisotopic peak signals over the peak elution period for spiked isotope-labeled internal standards and sample signals. Data processing was done using Agilent MassHunter Qualitative Analysis B.05.00. The single mass-to-charge ratio (m/z) expansion for the EIC was set as asymmetric with parameters set at -0.05 and $+0.1$ m/z for integration. The relative integrated abundances as observed for analytes and standards were used to calculate analyte concentrations based on the spike level and charge state; those values were then further averaged to obtain a single value for an analyte. Data were collected in profile mode over the range of 30 to 1700 m/z at a rate of one spectrum/second.

Pharmacokinetics of NJH395, its catabolite MIM697, and variants

To quantify the total ISAC (antibody conjugated with at least one TLR7 agonist) and total antibody (antibody with or without a TLR7 agonist) from plasma samples in non-GLP *in vivo* studies, we applied two GyroLab-based (Gyros Protein Technologies) immunoassays. The assays used an anti-human IgG capture reagent and either an orthogonal anti-human IgG or antipayload detection reagent respectively. The lower limit of quantification (LLOQ) was determined for each assay run individually based on the performance of the calibration curve using NJH395 and ranged between 0.0003 and 0.03 μ g/mL. Quantitative determination of the total antibody and total ISAC was assessed using two validated ELISA assays. The LLOQ of both assays was 0.093 μ g/mL. The quantitative analysis of the total antibody and total ISAC from the serum in the clinical phase I study was assessed using two validated ELISA assays. The LLOQ for the clinical total antibody assay and total ISAC assay was 0.075 μ g/mL.

A GyroLab-based immunoassay was applied to determine ADAs from plasma samples in non-GLP *in vivo* studies. The assay uses biotinylated NJH395 as a capture reagent and an anti-macaque IgG-specific detection reagent. ADAs against NJH395 were assessed in the serum samples using a validated homogeneous ELISA for the GLP toxicology study.

The concentration of the NJH395 catabolites MIM697 in the plasma was determined using high-performance LC with tandem MS (LC-MS/MS). The LLOQ for MIM697 in plasma was 0.3 to 2 ng/mL (for non-GLP studies) and was 0.3 ng/mL for both the GLP toxicity study in monkeys and the clinical phase I study.

Binding assays to human and cynomolgus monkey HER2

HER2 extracellular domains (ECD; Sino Biological, cat. #10004-H08H, 90295-C08H, and 80079-R08H) and ado-trastuzumab emtansine were reconstituted as directed by the manufacturers. All samples were diluted to working concentrations in running buffer (10 mmol/L HEPES, 150 mmol/L NaCl, pH 7.4). Antibodies were buffer-exchanged to the running buffer. Studies were performed on a Biacore 8K (GE Healthcare) using the parallel kinetics method and Fc capture of the antibodies. The chip was first preconditioned by cleaning a fresh CM5 chip using two 10-second pulses of 100 mmol/L HCl, two 10-second pulses of 50 mmol/L NaOH, and two 10-second pulses of 0.5% SDS. All 16 channels were activated using EDC/NHS per the manufacturer's instructions. Mouse antihuman immunoglobulin (Ig) G, Fc-specific capture antibody (Invitrogen, cat. #: 05-4200) was diluted to 2 µg/mL in 10 mmol/L sodium acetate, pH 5.0, and injected for 420 seconds at 10 µL/minute, followed by a 420-second pulse of 1 M ethanolamine. The 16 channels each contained $8,457 \pm 138$ RU of amine-coupled capture antibody. The same amount of NJH395 or ado-trastuzumab emtansine was captured on flow cell 2 of all eight channels (ligand), followed by the analyte HER2, where a different concentration was simultaneously injected over each channel. This allowed the full dose response to be acquired in a single injection, minimizing the number of cycles and the data acquisition time because of the long dissociation period. Human and rat HER2 ECDs were diluted from 100 to 0.78 nmol/L and cyno from 200 to 1.56 nmol/L—concentrations above 200 nmol/L showed significant nonspecific binding to the chip surface. The binding data were collected at 25°C with a 180-second contact time, followed by a 3600-second dissociation time in 10 mmol/L HEPES, 500 mmol/L NaCl, 3 mmol/L EDTA, 0.05% Tween 20, and pH 7.4 at a flow rate of 30 µL/minute. The surfaces were regenerated (antibody and remaining antibody/HER2 complex removed) using a 30-second pulse of 4 M MgCl₂, followed by a 30-second pulse of 10 mmol/L glycine, pH 2.0. The kinetic parameters were calculated using the vendor's 8K software.

HER2⁺ tumor cell binding and internalization of compounds

Binding of human antibodies and antibody conjugates on HCC1954 cells was measured by flow cytometry. HCC1954 cells were removed from expansion flasks with Accutase, counted, and resuspended to 1×10^6 cells/mL in cell culture media. 100 µL was aliquoted into wells of a 96-well U-bottom plate (Corning, #3799). Cells were pelleted by centrifugation and then media were removed. Test articles were serially diluted in cell staining buffer (BioLegend, #420201) and 100 µL was added to cells and incubated for 30 minutes on ice. Cells were pelleted by centrifugation and washed twice with 150 µL cell staining buffer. Secondary detection antibody, anti-human Fc-specific AF647 reagent (Jackson Immuno Research #709-606-98), was diluted 1/100 in cell staining buffer and 100 µL was added to wells and incubated for 30 minutes on ice. Cells were pelleted by centrifugation, washed, and then resuspended in 150 µL cell staining buffer containing 3 µmol/L DAPI and measured on an LSRFortessa.

To study the internalization of NJH395 on HCC1954 cells, the Zenon pHrodo iFL green system was used (Thermo Fisher #Z25611). Internalization was measured by flow cytometry at 0, 30, 120, 240, and 480 minutes. HCC1954 cells were removed from expansion flasks with Accutase, counted, and resuspended to 2×10^6 cells/mL in cell culture media. A 4× working solution of NJH395 and NJH397 was prepared in 1 mL cell culture media to yield a final concentration of 40 nmol/L when plated. 25 µL of the working solution for each antibody was aliquoted in triplicate in 96-well plates (Greiner Bio-

one#650161). One plate was prepared for each time point measured. 25 µL of cell culture media was added in triplicate for unstained control. A 4× working solution of Zenon pHrodo iFL labeling reagent was prepared and 25 µL was added to each well containing the 25 µL of antibodies or media added previously then incubated for 5 minutes at room temperature. 50 µL of cells previously prepared at 2×10^6 cells/mL were added to each well. Zero timepoint plate was placed on ice immediately, whereas additional plates were incubated for a time indicated at 37°C, 5% CO₂. After incubation, 100 µL of cold sodium azide containing cell staining buffer (BioLegend, #420201) was added to each well and then the plate was placed on ice until the final plate completed incubation. After incubation of all plates was completed, cells were pelleted by centrifugation, washed 2× with 150 µL cell staining buffer, and then suspended in 150 µL cell staining buffer containing 3 µmol/L DAPI and read on the LSRFortessa.

In vitro human peripheral blood mononuclear cell cytokine release assays

Fresh blood was obtained from healthy human donors through The Scripps Research Institute Normal Blood Donor Program. The blood was diluted in an equal volume of RPMI-1640 medium, then layered over a Ficoll-Plus cushion in LeucoSep tubes and centrifuged at $800 \times g$ for 15 minutes. The buffy coat layer was removed and added to an equal volume of RPMI-1640 medium and then centrifuged at $300 \times g$ to pellet the cells. Platelets were removed by sequential resuspension and centrifugation at low speed ($200 \times g$) in D-PBS, 5% HI-FBS, 1-mmol/L EDTA. After three rounds of platelet wash, the peripheral blood mononuclear cell (PBMC) pellets were resuspended in RPMI-1640 with GlutaMAX, 5% HI-FBS, 10-mmol/L HEPES, 0.05-mmol/L β-Mercaptoethanol, and 100 U/mL penicillin-streptomycin (1×). Cells were counted and plated in 30 µL/well in 384-well tissue culture plates at 50,000 (IL6 readout) or 150,000 (IFNα readout) cells/well. Test compounds were serially diluted in an assay medium with a constant concentration of dimethyl sulfoxide (DMSO) and 10 µL added to each test well. After overnight incubation, 6 µL of cell supernatant was sampled from each well for each assay. IFNα was measured using the human IFNα2b AlphaLISA assay on an EnSpire Plate Reader (PerkinElmer), and IL6 was measured using a human IL6 HTRF kit (CisBio) and read on a PHERAstar Plate Reader (BMG Labtech). Resiquimod was used as the reference compound for measuring the percent efficacy in both assays.

In vitro human whole blood cytokine release assays

In vitro cytokine release experiments were conducted in human whole blood obtained from 8 healthy donors through the NIBR campus donor program (Basel, CH) incubated with an anti-HER2 ISAC (NJH395), a control ISAC (NJH397), and the active TLR7 catabolite MIM697, and the anti-HER2 antibody MIW338. In parallel, lipopolysaccharide, alemtuzumab, and the specific TLR7 agonist ssRNA/DOTAP were included as positive controls to validate the functionality of the assay system and act as clinical benchmarks. After 6 and 24 hours, plasma supernatants were collected, and the cytokines were determined using a bead-based multiplex immunoassay approach (Luminex-based detection kit for multiplexed determination) using a Human Cytokine Magnetic Bead Panel, Milliplex MAP Kit (Merck Millipore-Milliplex). Next, test items were included at concentrations expected to correspond to the expected C_{max} concentrations attained after intravenous administration of NJH395 in the first-in-human clinical trials.

Target gene expression via quantitative PCR

Whole blood was obtained from 3 healthy donors from the NIBR blood donor program (run by Medco onsite in Cambridge, MA) and collected in sodium heparin tubes. Blood was diluted 1:1 in RPMI media (Gibco), and 1 mL/well was cultured in 6-well tissue culture plates. NJH395 or naked anti-HER2 (MIW338) was added at varying final concentrations between 0.04 and 5 mg/mL, and samples were cultured at 37°C, 5% CO₂, for 3 days. After 3 days, cultures were transferred to sterile 15-mL conical tubes (Falcon) and centrifuged (300 × *g* for 5 minutes at 4°C). The supernatant was removed, and RBC lysis and RNA extraction were performed using the QIAamp RNA Blood Mini Kit (#52304; Qiagen). The RNA was quantified on a NanoDrop 1000 instrument (Thermo Fisher) and stored at –80°C. Reverse transcription of the RNA was done using the High-Capacity cDNA Reverse Transcription Kit (Applied Biosystems, Hi-Capacity cDNA reverse transcription kits; #4368813). 0.5–1.0 µg RNA was used for 50 µL RT reactions. RT reaction (4.5 µL) was used for quantitative (q)PCR using the TaqMan Fast Advanced Master Mix (#4444557; ABI). qPCR was done using Best Coverage primers from Applied Biosystems [Hs02786624_g1 for GAPDH (for normalization), Hs00973635_m1 for OAS1, Hs00942643_m1 for OAS2, Hs00174131_m1 for IL6, Hs00174128_m1 for TNFα, and Hs01555410_m1 for IL1β]. qPCR reactions were read on the Applied Biosystems ViiA 7 system. Reactions were run with 3 biological replicates (3 blood donor samples) per treatment group. Gene expression was determined using the 2^{–ΔΔCT} method. Data were plotted as fold change with each antibody compared with PBS control reaction.

In vitro reporter assays

HEK293T cell lines engineered to stably coexpress luciferase under the control of the NF-κB-responsive promoter and either the human Toll-like receptor 7 (TLR7; HEK293T-TLR7-NF-κB-luc) or the human TLR8 (HEK293T-TLR8-NF-κB-luc) were used to characterize agonist compounds. On the day before compound treatment, media were removed from the T175 flasks, and cells were trypsinized and resuspended in DMEM supplemented with 2-mmol/L glutamine, 3% FBS, and 100 U/mL of penicillin–streptomycin (1×) and plated at 25,000 cells/well in 40 µL/well. Assay plates were incubated at 37°C overnight. On the day of the assay, the serially diluted compounds were added to the assay test wells at 10 µL/well. Reference compounds resiquimod (InvivoGen) and motolimod (APEX-BIO) and the negative control, DMSO (also added at 5× dilution) were included in each assay. After a 16-hour incubation, half the media were removed, and 25 µL of undiluted Bright-Glo reagent (Promega) was added per well. Plates were gently mixed, and the luciferase activity was measured on a CLIPR luminescence plate reader (Molecular Devices Corporation). Data were acquired using the embedded SoftMaxPro software and analyzed by a curve-fitting program implemented using Matlab version R2009a. EC₅₀ values were calculated by performing logistic regression on measured dose–response data points using a four-parameter, variable slope equation.

Analysis of THP-1 cells and HER2-expressing cancer cell cocultures

A THP-1 cell line was subcultured every 2 to 4 days, was not allowed to exceed 1 × 10⁶ cells/mL, and used at passages 5 to 12 for the coculture assay. THP-1 cells were plated at 25,000 cells/well in TC-treated 96-well flat-bottomed plates (#353072; Falcon) in 200 µL growth media as described above with 125 ng/mL phorbol-12-myristate-13-acetate (PMA; #524400; Sigma) and then incubated for 48 hours at 37°C,

5% CO₂, to generate adherent macrophage-like cells. Cells were washed with PBS to remove PMA. HCC1954 cells were removed from expansion flasks with Accutase, counted, and resuspended to 0.25 × 10⁶ cells/mL in growth media. Next, 100 µL was added to the wells containing adherent THP-1 cells. Test articles were serially diluted in growth media, and 100 µL was added to test wells. Plates were incubated for 48 hours. Cells were pelleted by gentle centrifugation, and then 50 µL of supernatant from each well was transferred to a 96-well U-bottomed storage plate (#650161; Greiner Bio-one) and placed at –80°C.

Cytokine analysis

To measure cytokines, the Human V-PLEX Viral Panel 1 Kit (#K15345D-2; Meso Scale Discovery) was used. Plates were washed three times with 1× wash buffer (20× KPL Wash Solution, Sera Care; #5150-0011, diluted to 1× in DI water, Invitrogen; #10977-015). Supernatant samples were diluted 1:5 in Diluent 2 and added to plates. Calibrator for standard curve generation was provided in the kit and prepared following the manufacturer's recommendations. Plates were treated according to kit directions and analyzed on the SECTOR Imager 6000 (MSD Instrument). Results were analyzed and concentrations were generated with Discovery Workbench Software Version 4.0 (Mesoscale Discovery).

Flow cytometry

HER2 was measured by flow cytometry using detection reagent clone 24D2 (#324408; BioLegend), and cells from the coculture assays above were analyzed. Following the removal of supernatants for cytokine evaluation from the 48-hour coculture, adherent cells were washed with 150 µL PBS, then 150 µL Accutase was applied to each well and incubated at 37°C for 10 minutes. Cells were transferred to a U-bottomed 96-well assay plate, washed with PBS, and stained for 10 minutes at room temperature with 100 µL of live/dead stain (1/1,000 dilution; #423108; BioLegend). Cells were Fc-blocked for 10 minutes on ice with 5 µg/mL Fc block (#422302; BioLegend). Cells were then stained for 30 minutes on ice with 100 µL of antibody cocktail prepared in the cell staining buffer (#420201; BioLegend) containing anti-human CD45 (#563792; BioLegend) and anti-human HER2 clone described above. The fluorescence intensity of HER2 on CD45⁺ THP-1 cells and CD45[–] HCC1954 cells was measured on an LSRFortessa.

Intracellular analysis

Intracellular phospho-NF-κB was measured in THP-1 cells 1 hour after treatment with MIM697. THP-1 cells were plated at 40,000 cells/well in TC-treated 96-well flat-bottomed plates (#353072; Falcon) in 200-µL growth media with 125-ng/mL PMA (#524400; Sigma) and then incubated for 48 hours at 37°C, 5% CO₂, to generate adherent macrophage-like cells. Media were removed and replaced with 200 µL fresh growth media without PMA, and cells were rested for 24 hours at 37°C, 5% CO₂. Media were removed, and 50 µL fresh growth media were added to cells. MIM697 was serially diluted in growth media, and 50 µL was added to test wells. The plate was incubated for 1 hour at 37°C, 5% CO₂. To measure NF-κB, Thunder Phospho-NF-κB (S536) TR-FRET Cell Signaling Assay Kit (#KIT-NFKBP-100; BioAuxilium Research) was used. Cells were treated according to kit directions for the Standard 2-Plate Assay Protocol for Adherent Cells. Lysates were transferred to a 384-well proxi-plate (#6008280; PerkinElmer) and incubated overnight at 4°C with a detection antibody. TR-FRET signal was recorded at 620 and 665 nm on EnVision.

qPCR

qPCR was used to determine the TLR7-negative status of HCC1954 cells. HCC1954 cells were cultured as described above. HEK293T cells engineered to overexpress human TLR7 (reporter assay described above) were used as a positive control for the assay and cultured in DMEM supplemented with 10% FBS, 1% penicillin-streptomycin, 5 mg/mL puromycin, and 5 mg/mL blasticidin. Cells were plated at 15,000 cells/well in a 384-well plate (#789063; Greiner Bio-one) and incubated overnight at 37°C in 5% CO₂. Cells were harvested at 24 hours using TaqMan Gene-Expression Cells-to-CT (#AM1728; Thermo Fisher Scientific), and qPCR was performed per the manufacturer's protocol. TLR7 primer HS01933259_s1 (#4331182; Thermo Fisher Scientific) and Human GAPD (GAPDH) Endogenous Control (VIC/MGB; #4326317E; Thermo Fisher Scientific) were used. Data were acquired using a QuantStudio6 and analyzed using GraphPad/Prism software.

Preclinical xenograft studies

All animal-related procedures were conducted under a Genomics Institute of the Novartis Research Foundation Institutional Animal Care and Use Committee (GNF IACUC)-approved protocol in compliance with Animal Welfare Act regulations and the Guide for the Care and Use of Laboratory Animals. SCID beige and Foxn1 nude (Envigo) female mice (ages 6–8 weeks) were used as the experimental animals. For implantation, cells were resuspended in FBS-free media and Matrigel (Corning, cat. #47743–706) at 1:1, and 1×10^6 (N87) or 5×10^6 (HCC1954 and SKOV-3) cells were implanted subcutaneously in the lower flank (N87 and SKOV-3) or orthotopically into the mammary fat pad (HCC1954; 100 μ L injection volume). Tumor sizes were assessed three times a week once tumors were palpable. Tumor sizes were determined using caliper measurements. Tumor volumes were calculated using the following formula: (length $\times$ width $\times$ width)/2. The mice would be sacrificed as tumor volume over 2,000 mm³ or mouse was moribund for survival. Before dosing, animals were randomized into groups ($n = 8$) based on the tumor size. MIW338 (anti-HER2 unconjugated) was formulated at 5 mg/kg, NJH397 (isotype antibody MHW079 conjugate) at 5 mg/kg, and NJH395 at 1 mg/kg, 3 mg/kg, and 5 mg/kg in PBS, and was administered in one single dose intravenously. Clodronate liposomes were dosed at 0.7 mg/kg IP 4 days prior to antibody dosing and then on study days 2 and 6.

For IHC samples, tumors were collected at day 0 (predose) or day 5 relative to NJH395 dose from 2 animals per group per timepoint and fixed in formalin. Paraffin-embedded tumors were then sliced and analyzed as described below. All data are expressed as mean $\pm$ SEM. Between-group comparisons were performed in Prism (GraphPad) using a one-way ANOVA followed by a *post hoc* Sidak's multiple comparisons test.

IHC of mouse tumor tissues

IHC staining for CD3 [cat. #790-441; Ventana Medical Systems (VMS)], Iba1 (cat. #019-19741; Fujifilm Wako Pure Chemical Company), CD68 (cat. #790-2931; VMS), CD163 (cat. #760-4437; VMS), and GFAP (cat. #7604345; VMS), including the deparaffinization and antigen-retrieval steps, was performed on a Ventana Discovery XT autostainer using standard Ventana Discovery XT reagents (VMS). Slides were deparaffinized and then submitted to heat-induced antigen retrieval by covering them with Cell Conditioning 1 (CC1/pH8) solution per the standard Ventana retrieval protocol. Then, slides were incubated with the primary antibodies or a nonimmune isotype-matched control for 1 hour. Visualization was performed

by incubation with the Ventana Discovery OmniMap HRP reagent, followed by Ventana Discovery ChromoMap 3,3'-diaminobenzidine (DAB). Furthermore, counterstaining was performed using Ventana Hematoxylin and Ventana Bluing reagent for 4 minutes each. Slides were dehydrated, cleared, and coverslipped with a synthetic mounting medium.

NJH395 toxicity in a nonhuman primate model

NJH395 pharmacology, pharmacokinetics, and toxicity were assessed in single- and repeat-dose studies using cynomolgus monkeys. Primate study protocols (Novartis study 1570283 and Novartis study 1570362) were reviewed and approved by the Institutional Animal Care and Use Committee of Novartis Pharmaceuticals and were conducted at Novartis Pharmaceuticals and Charles River Laboratories; conformed to the Guide for the Use and Care of Laboratory Animals (National Research Council of the National Academies, Institute for Laboratory Animal Research, 2011); and were based on requirements of the U.S. Department of Health and Human Services, Food and Drug Administration (FDA), United States Code of Federal Regulations, Title 21, Part 58: Good Laboratory Practice (GLP) for the Nonclinical Laboratory Studies and as accepted by the Regulatory Authorities throughout the European Union (OECD Principles of GLP) and Japan (MHLW). Animals were pair- or group-housed ($n = 2$ –3/cage; same-sex and same dosing group together) by treatment group in a 12-hour light/dark cycle facility in stainless-steel modified quad cages equipped with a stainless-steel mesh floor and an automatic watering valve. Water and food were provided *ad libitum*. The total number of animals used in these studies, as well as the group size and the number of groups, were considered to be the minimum required to properly characterize the effects of NJH395. Descriptive statistics (means and SDs) are presented for drug exposure and antidrug antibodies (ADA). Individual animal data are plotted for cytokine assays.

Cynomolgus monkeys (*Macaca fascicularis*) were randomly assigned to treatment groups using a computer-based procedure (male and female monkeys randomized separately), and the vehicle, NJH395, or ISAC isotype control NJH397 were administered intravenously (dose volume: 2 mL/kg) by slow bolus injection into a suitable peripheral vein. The monkeys were obtained from either Worldwide Primates (Asian mainland; study 1570283) or Charles River Laboratories (Chinese origin; study 1570362), and at the initiation of dosing, the monkeys were aged approximately 2 to 4 years and weighed approximately 2 to 4 kg. Animal health was monitored for general health by clinical observations (daily), body weight (weekly), food consumption (qualitative; once daily), body temperature (digital infrared or digital rectal thermometer; predose and 6 hours post-dose), respiration rate (visual assessment; pretreatment and 3–6 hours post last dose administration), clinical pathology (femoral vein; hematology, coagulation, clinical chemistry, urinalysis; pretreatment, predose, postdose).

In an 11-week investigative intravenous pharmacokinetic/pharmacodynamic and toxicity study (study 1570283; IACUC protocol TX-4039), male monkeys were administered NJH395 in PBS (16.2 mg/mL) at 3 mg/kg (days 1 and 64; $n = 4$ male monkeys), 10 mg/kg (days 1, 50, and 85; $n = 4$ male monkeys), or 30 mg/kg (single doses; $n = 1$), and an additional group was administered the isotype control NJH397 in PBS (14.2 mg/mL) at 10 mg/kg (days 1 and 64; $n = 2$ male monkeys). Blood samples were collected by venipuncture following each dose to analyze the pharmacokinetics and immunogenicity of NJH395 and catabolic breakdown product MIM697 (following administration of two doses at the 3 mg/kg dose level at days 1 and

64; three doses of 10 mg/kg dose levels at days 1, 50, and 85; after a single dose of 30 mg/kg; at day 85); and pharmacokinetics of isotype control NJH397 (following administration of 10 mg/kg on days 1 and 64). Blood sample time points for pharmacokinetics and immunogenicity were predose and at 1, 6, 24, 48, 96, 168, 240, 336, 408, 504, and 600 hours after each dose; and for serum cytokines (serum separator tubes)(immediately predose and 1, 3, 6, 24, and 48 hours post each dose).

In a 7-week single or repeated intravenous dose study (study 1570362; a GLP study), male and female monkeys were administered vehicle control or NJH395 (lyophilized NJH395 was reconstituted in sterile water for injection, USP, and then formulated in 20 mmol/L histidine [His], 240 mmol/L sucrose, 0.02% polysorbate 20, and pH 5.7), via an intravenous (slow bolus) injection either once (3 and 10 mg/kg/dose; $n = 5/\text{sex}/\text{group}$) on day 1 or three times (0 mg/kg/dose; 3/sex or 1 mg/kg/dose; 5/sex) on days 1, 22, and 43 (total of three doses). Blood samples were collected by venipuncture following each dose to analyze the total ISAC and total antibody in the serum for NJH395 and metabolite MIM697 in serum [serum separator (for IG and TKS = TK for total Ab and total ISAC)] or plasma (K2EDTA for TKP = TK for small molecule; days 1, 22, 43, 64; predose, 2, 6, 24, 96, 168, 504 hours postdose) and for serum cytokines (serum separator tubes; days 1, 22; predose, 2, 6, 12, 24, 96, 504 hours postdose; day 43: 2, 6, 12, 24, 96 hours postdose).

In the 7-week study, animals were euthanized at the end of the dosing period, except for a single 30 mg/kg animal, which was euthanized on day 8 owing to adverse clinical signs, including decreased activity. We conducted gross pathology examination on the terminal cases and all unscheduled deaths, as well as microscopic examinations (formalin-fixed, hematoxylin and eosin stained) of tissues from a select list of organs and tissues [including aorta, bone marrow smear, bone marrow, sternum bone, brain, cervix, epididymis, esophagus, eye, gallbladder, adrenal gland, mammary gland, parathyroid gland, pituitary gland, prostate gland, salivary gland, seminal vesicle, thyroid gland, gut-associated lymphoid tissue, heart, kidney, cecum, colon, rectum, liver, lung, lymph nodes (mandibular, mesenteric, retropharyngeal), skeletal muscle, optic, and sciatic nerves, ovary, pancreas, dose administration site, skin, duodenum, ileum, jejunum, spinal cord, spleen, stomach, testis, thymus, tongue, trachea, urinary bladder, uterus, vagina] including gross lesions. Furthermore, histopathologic assessments were performed on routine formalin-fixed, paraffin-embedded sections, and IHC analyses and *in situ* hybridization were performed on the control and the single 30 mg/kg animal that was euthanized on day 8.

Primate cytokine analysis

Serum samples from monkey blood collected in the 7-week and 11-week studies (described above) were stored at $\leq -70^\circ\text{C}$ until analysis. Samples were assessed with the Cytokine Monkey Magnetic 29-Plex Panel (Thermo Fisher Scientific, Invitrogen, LPC0005M) using the Luminex xMAP platform. Kit analytes assessed were EGF, Eotaxin, G-CSF, GM-CSF, IL1 β , IL-1RA, IL2, IL4, IL5, IL6, IL8, IL10, IL12p70, IL15, IL17, IFN γ IP-10, MCP-1, MDC, MIF, MIG, MIP1 α , MIP1 β , I-TAC, RANTES, TNF α , and VEGF. Analysis of data was based on the fluorescence intensity minus background (FI-Bkgd) extrapolation from a five-parameter (5-PL) curve-fitting program as a part of the Bio-Plex manager Software 6.1 analysis platform as a part of the Bio-Plex-200. Of note, values were evaluated relative to pretest concentrations.

Pathology and *in situ* hybridization of nonhuman primate tissues

Molecular localizations (IHC and *in situ* hybridization) studies were performed to characterize inflammatory infiltrates in the brain of the

early decedent nonhuman primate dosed with 30 mg/kg of NJH395 euthanized at 8 days and compared with an archived control animal. IHC staining for CD3 (IgG1 clone 2GV6 prediluted; cat. #790-4341; VMS), Iba1 (rabbit polyclonal at 1 $\mu\text{g}/\text{mL}$; cat. #019-19741; Fujifilm Wako Pure Chemical Company), CD68 (IgG1/kappa clone KP1 prediluted; cat. #790-2931; VMS), CD163 (IgG1 clone 10D6 prediluted; cat. #760-4437; VMS), and GFAP (prediluted; cat. #7604345; VMS), including the deparaffinization and antigen-retrieval steps, was performed on a Ventana Discovery XT autostainer using standard Ventana Discovery XT reagents (VMS). Slides were deparaffinized and then submitted to heat-induced antigen retrieval by covering them with Cell Conditioning 1 (CC1/pH8) solution per the standard Ventana retrieval protocol. Then, slides were incubated with the primary antibodies or a nonimmune isotype-matched control for 1 hour. Visualization was performed by incubation with the Ventana Discovery OmniMap HRP reagent, followed by Ventana Discovery ChromoMap 3,3'-diaminobenzidine (DAB). Counterstaining was performed using Ventana Hematoxylin and Ventana Bluing reagent for 4 minutes each. Slides were dehydrated, cleared, and coverslipped with a synthetic mounting medium.

We performed *in situ* hybridization to detect Mfa-IL1b (cat. #433818; Advanced Cell Diagnostics [ACD]), Mfa-IL6 (cat. #533839; ACD), Mmu-PPIB (positive control and tissue quality control; cat. #457719; ACD), and DAPB (negative control; cat. #312039; ACD) transcripts using reagents and equipment supplied by the ACD and VMS. Of note, ACD designed the *in situ* hybridization RNAscope probes. Positive peptidylprolyl isomerase B (PPIB) and negative dihydronicotinamide reductase (DAPB) control probe sets were included to ensure the mRNA quality and specificity, respectively (data not shown). The hybridization method followed protocols established by ACD and Ventana systems using a DAB chromogen. Briefly, 5-mm sections were baked at 60°C for 60 minutes and used for hybridization. The deparaffinization and rehydration protocol was performed using a Sakura Tissue-Tek DR5 stainer with the following steps: three times xylene for 3 minutes each, two times 100% alcohol for 3 minutes, and air-dried for 5 minutes. Offline manual pretreatment in $1\times$ retrieval buffer at 98°C to 104°C for 15 minutes for all tissue sections. Next, optimization was performed by first evaluating PPIB and DAPB hybridization signals and subsequently using the same conditions for all slides. Following pretreatment, the slides were transferred to a Ventana Ultra autostainer to complete the hybridization procedure, including protease pretreatment, hybridization at 43°C for 2 hours, followed by amplification, and detection with horseradish peroxidase and hematoxylin counterstain.

H&E-stained sections and molecular localization slides were reviewed by veterinary pathologists on an Olympus BX46 microscope and scanned on an Aperio AT2 scanner (Leica Biosystems) at $20\times$ for representative photomicrographs.

ADA binding via surface plasmon resonance

HER2 ECDs (cat. #10004-H08H, 90295-C08H, and 80079-R08H; Sino Biological) and ado-trastuzumab emtansine were reconstituted per the manufacturers' instructions. All samples were diluted to working concentrations in running buffer (10 mmol/L HEPES, 150 mmol/L NaCl, pH 7.4). Antibodies were buffer-exchanged to the running buffer. Studies were performed on a Biacore 8K (GE Healthcare) using the parallel kinetics method and Fc capture of the antibodies. The chip was first preconditioned by cleaning a fresh CM5 chip using two 10-second pulses of 100 mmol/L HCl, two 10-second pulses of 50 mmol/L NaOH, and two 10-second pulses of 0.5% SDS. All 16 channels were activated using EDC/NHS per the manufacturer's instructions. Mouse anti-human immunoglobulin (Ig) G, Fc-specific

capture antibody (cat. #05-4200; Invitrogen) was diluted to 2 mg/mL in 10-mmol/L sodium acetate, pH 5.0, and injected for 420 seconds at 10 mL/min, followed by a 420-second pulse of 1-M ethanolamine. Each of the 16 channels contained $8,457 \pm 138$ RU of amine-coupled capture antibody. The same amount of NJH395 or ado-trastuzumab emtansine was captured on flow cell 2 of all eight channels (ligand), followed by the analyte HER2, where a different concentration was simultaneously injected over each channel; this allowed obtaining the full dose response in a single injection, thereby minimizing the number of cycles and the data acquisition time because of the long dissociation period. Human and rat HER2 ECDs were diluted from 100 to 0.78 nmol/L and cyno from 200 to 1.56 nmol/L—concentrations above 200 nmol/L exhibited significant nonspecific binding to the chip surface. The binding data were collected at 25°C with a 180-second contact time, followed by a 3,600-second dissociation time in 10 mmol/L HEPES, 500 mmol/L NaCl, 3 mmol/L EDTA, 0.05% Tween 20, and pH 7.4 at a flow rate of 30 mL/minute. The surfaces were regenerated (antibody and remaining antibody/HER2 complex removed) using a 30-second pulse of 4-M MgCl₂, followed by a 30-second pulse of 10 mmol/L glycine, pH 2.0. The kinetic parameters were calculated using the vendor's 8K software.

Cytokine determination in human whole blood from patients in the phase I study

Blood samples were collected from patients at C1D1 predose, C1D1 6 hours postdose, C1D5, C1D15, and unscheduled visits, as applicable. Only the first two time points were used for this analysis. Blood (3 mL) was collected in K₂EDTA tubes and centrifuged at 2000 RCF for 25 minutes at 4°C. Clear plasma was collected, aliquoted, and stored at -70°C. Cytokine concentration in patient plasma was assessed at BioAgilytix using the Meso Scale Diagnostics platform according to the manufacturer's instructions using catalog items K15049D, K15049G, C4049, K15047D, K15047G, and C4047. Plasma sample (25 µL) was diluted 1:1 into sample diluent buffer and 50 µL was added to the plate. Standard curves were generated with calibrators contained in each kit.

Luminescence was read using the Sector S 600 plate reader. Luminescence data were imported into Gen5 v2.01 software (BioTek Instruments), log transformed, and plotted using a 4PL curve fit without weighting. Only readouts with a detectable change in levels were reported.

Clinical study design and patient population

A phase I, multicenter, open-label study (NCT03696771) was conducted in four countries per the International Council on Harmonization Guidelines for Good Clinical Practice, applicable local regulations, and the ethical principles listed in the Declaration of Helsinki. Male or female patients (age ≥ 18 years with histologically or cytologically confirmed/documented HER2⁺ solid tumors (defined as IHC-3⁺ or FISH-amplified), excluding breast cancer, who had progressed or were intolerant to all approved therapies known to confer clinical benefit were eligible. The key eligibility criteria included Eastern Cooperative Oncology Group performance status ≤ 2 , measurable disease per Response Evaluation Criteria in Solid Tumors (RECIST) v1.1, and the absence of symptomatic brain metastases. Patients previously treated with a TLR7/8 agonist, patients with autoimmune disease, and/or who discontinued prior immune-checkpoint inhibitor (ICI) treatment due to toxicity were excluded. All patients provided written informed consent before participation in the study.

The initial phase I study design included a single ascending dose (SAD) part, followed by a multiple ascending dose part. The number of patients included in each group is included in **Table 1**. The primary objective was to characterize the safety and tolerability of NJH395 as monotherapy. The key secondary objectives included characterization of NJH395 pharmacokinetics (PK) and its catabolite MIM697, presence of NJH395 antibodies, and assessment of preliminary antitumor efficacy, as detailed. Patients received NJH395 once in the SAD part, and the incidence of dose-limiting toxicities (DLT) was assessed in the first 21 days. An adaptive Bayesian logistic regression model guided by the escalation with the overdose control principle

Table 1. Baseline demographic and clinical characteristics of patients treated with a SAD of NJH395.

Demographic variables	NJH395 dosing cohorts for SAD in non-breast HER2 ⁺ tumors ^a					All patients N = 18
	0.1 mg/kg n = 4	0.2 mg/kg n = 3	0.4 mg/kg n = 4	0.8 mg/kg n = 3	1.6 mg/kg n = 4	
Age, years, median (range)	53.5 (42–74)	51.0 (46–59)	53.5 (51–72)	53.0 (48–56)	51.0 (45–72)	52.5 (42–74)
Sex, n (%)						
Male	3 (75.0)	2 (66.7)	3 (75.0)	1 (33.3)	1 (25.0)	10 (55.6)
Female	1 (25.0)	1 (33.3)	1 (25.0)	2 (66.7)	3 (75.0)	8 (44.4)
ECOG PS, n (%)						
0	2 (50.0)	1 (33.3)	2 (50.0)	2 (66.7)	1 (25.0)	8 (44.4)
1	2 (50.0)	2 (66.7)	2 (50.0)	1 (33.3)	3 (75.0)	10 (55.6)
Cancer diagnosis, n (%)						
Colorectal	2 (50.0)	1 (33.3)	2 (50.0)	3 (100.0)	3 (75.0)	11 (61.1)
Bladder			1 (25.0)			1 (5.6)
Esophageal		1 (33.3)				1 (5.6)
Gastric cancer	1 (25.0)					1 (5.6)
Nasopharyngeal		1 (33.3)				1 (5.6)
NSCLC	1 (25.0)					1 (5.6)
Pancreatic			1 (25.0)			1 (5.6)
Small Intestine					1 (25.0)	1 (5.6)
Number of prior antineoplastic medication regimen						
Median	6	3	2	5	3.5	3.5

Abbreviations: ECOG PS, Eastern Cooperative Oncology Group performance status; FU, follow-up; NSCLC, non-small cell lung cancer.

^aTreatment period, 1 cycle 21 days; end of treatment, day 22; follow-up period, day 30, 43, 90, 150, and 196 of safety FU.

(performed by R package OncoBayes2 authorized by Novartis) was used to guide dose escalation. The starting dose level was 0.1 mg/kg, and pharmacodynamic/biomarker assessment was performed on predose (screening) and postdose (day 5 $\pm$ 3 days) tumor biopsies. Biomarkers were analyzed using IHC and RNA sequencing (RNA-seq) as described below.

NJH395 antitumor efficacy and safety in patients

All patients who received study treatment were included in the safety analysis. Adverse events (AE) were assessed and graded using the Common Terminology Criteria for Adverse Events version 5.0 (51). Tumor response was determined at the end of the treatment visit on day 22 + 7 days per local investigator's assessment using RECIST version 1.1 and immune-related RECIST (iRECIST; refs. 52, 53). For patients who discontinued treatment for reasons other than documented disease progression, lost to follow-up, or withdrawal of consent, tumor assessments must continue to be performed every 8 weeks ($\pm$ 7 days) until documented disease progression per investigator and iRECIST, death, lost to follow-up, or withdrawal of consent.

Determination of ADAs in patients and competitive binding assays

A three-tiered (screening, confirmation, and titration) immunogenicity evaluation was used to evaluate ADAs from the clinical study. A standard bridging ELISA format was used to detect the NJH395 antibodies. Samples confirmed positive for antibodies against NJH395 were further assessed for the presence of neutralizing antibodies to NJH395 function. Therefore, we used a competitive ligand binding assay using NJH395 and recombinant Erbb2/HER2. Samples confirmed positive for NJH395 antibodies were also evaluated for trastuzumab reactivity using excess trastuzumab. If positive, the samples were assessed in a neutralization assay for trastuzumab using trastuzumab and recombinant HER2/Erbb2.

IHC of clinical samples

TLR7

Formalin-fixed paraffin-embedded (FFPE) 4- μ m sections were received for 14 patients. TLR7 IHC staining was performed using the TLR7 antibody (rabbit monoclonal, anti-TLR7; clone EPR2008(2); Abcam ab124928, dilution 1:2,400) on the Roche Ventana Ultra Benchmark platform with Ventana UltraView Detection Kit. Sample sections were pretreated with cell conditioning solution (Ventana, CC1, #950-224) for 64 minutes, followed by primary antibody incubation for 32 minutes without heat and Ventana UltraView DAB detection. Run controls included tonsil positive control and no primary antibody control. Stained slides were scanned using the Leica Aperio Scanscope (Leica Biosystems). The images were imported into the HALO software ver. 3.0.311.261 (Indica Labs). Annotations were then drawn by a board-certified pathologist (SC) to exclude areas of necrosis, debris, or artifact. The area quantification was performed using the Area Quantification module ver. 2.1.3 [Stain RGB color settings (0.342, 0.498, 0.636); min OD 0.188]. TLR7 scoring was recorded as a percent positive tissue.

Payload and HER2

FFPE unstained slides for 14 patients (paired biopsies) and correlative metadata were received from Oncology Translational Research, Novartis Institutes for BioMedical Research, Cambridge, and IHC stains for HER2 and payload (TLR7 agonist) were performed. HCC1954 breast cancer cell line with and without payload treatment was used as positive and negative controls for the payload IHC assay.

IHC assays were performed on the BenchMark Ultra automated staining platform (Roche/VMS). Slides were deparaffinized using the Discovery Wash Solution (950-510; VMS). Antigen retrieval was then performed using the Ultra CC1 solution (950-224; VMS) at a high temperature (100°C) for 40 minutes. The rabbit polyclonal primary antibody LWZ685-TLR7 agonist (GNF) was diluted to 2 mg/mL in Ventana Discovery antibody diluent with pH 7.2 (760-108; VMS) and placed into a user-fillable dispenser. The HER2 antibody (4290s; Cell Signaling Technology) was diluted at 1:100 in the Ventana Discovery antibody diluent and placed into a user-fillable dispenser. Each of the primary antibodies was incubated for 1 hour at 37°C. The secondary antibody Discovery OmniMap anti-Rabbit HRP (760-4311; VMS) was then applied, and incubation for 12 minutes was performed. The signal was subsequently detected using the ChromoMap DAB Detection System (760-159; VMS) per the manufacturer's instructions. Counterstaining was completed with 25% of Mayer's modified hematoxylin (HXMMHGAL; American MasterTech) for 60 seconds, followed by a bluing/rinse step in tap water for 60 seconds. Slides were digitized using a Hamamatsu Nanozoomer s360 slide scanner at a magnification of 40 $\times$. The scanned slides were reviewed by a board-certified pathologist (LI) with the Leica ImageScope Viewer ($\times$ 64) and evaluated for payload staining. Visual analysis was used to assess payload staining in each cycle 1 day 5 (C1D5) biopsy as average intensity and the average percent of tumor staining. HER2 IHC stains and scoring were performed at an outside laboratory (HistoGeneX), and those HER2 scores were used in the analysis.

CD8, CD68, and PD-L1

Staining was performed at Cell Carta (formerly Histogenex) using 4- μ mFFPE sections. CD8 IHC (Clone C8/144B, cat. #M710301, Agilent Technologies, 1:75) and CD68 IHC (Clone KP-1, cat. #790-2931/05278252001; Roche VMS; ready to use) were performed on the Roche Ventana Benchmark XT platform (Roche VMS). PD-L1 IHC (Clone 22C3, cat. #SK00621; Agilent Technologies; pharmDx kit) was performed on Autostainer Link 48 using the PD-L1 IHC 22C3 pharmDx kit (Cat.#SK00621-5; Agilent Technologies). CD8 and CD68 slides were scanned using Panoramic scanning devices (3DHitech) and analyzed with the Definiens software (Definiens Inc) by a Cell Carta imaging scientist or anatomic pathologist to measure the percent marker area. PD-L1 scoring was performed for tumor percent positivity by a Cell Carta pathologist according to Agilent Technologies interpretation guidance for FDA-approved *in vitro* diagnostic use.

Tumor biopsy RNA-seq

Fresh tumor biopsies were collected with 3 to 6 needle passes at screening and days 5 to 8 postdose. Tissue was placed in 10% neutral formalin solution and fixed at room temperature for 6 to 24 hours. After fixation, the samples were transferred to 70% ethanol for storage until subsequent paraffin embedding. Total RNA was extracted from FFPE slides using the allPrep RNA Extraction from FFPE Tissue Kit (Qiagen). Ribosomal RNA from extracted total RNA was depleted using RNaseH (Sigma-327 Aldrich). The rRNA-depleted sample was then fragmented, converted to complementary DNA, and carried through the remaining steps of next-generation sequencing library construction—end repair, A-tailing, indexed adapter ligation, and PCR amplification—using the TruSeq RNA v.2 Library Preparation kit (Illumina). The captured library was pooled with other libraries, each having a unique adapter index sequence, and applied to a sequencing flow cell. The flow cell underwent cluster amplification and massively parallel sequencing by synthesis using Illumina v.4 chemistry and paired-end 100-bp reads (Illumina).

Whole blood RNA-seq

Fresh blood (2.5 mL) was collected in PAXgene Blood RNA tube at predose and at days 5 to 8 postdose, to match the biopsy collection. Samples were then stored at -80°C freezer until RNA isolation. Total RNA was extracted from whole blood using an extraction kit compatible with the specimen type (e.g., PaxGene, frozen). Alpha and beta-globin mRNA was depleted from the sample by using GLOBINclear (Invitrogen). The globin-depleted RNA was enriched for mRNA using poly-T probes binding to the mRNA's poly-A tail. The enriched mRNA was then fragmented, converted to complementary DNA, and carried through the remaining steps of next-generation sequencing library construction—end repair, A-tailing, indexed adapter ligation, and PCR amplification—using the TruSeq RNA v.2 Library Preparation kit (Illumina). The captured library was pooled with other libraries, each having a unique adapter index sequence, and applied to a sequencing flow cell. The flow cell underwent cluster amplification and massively parallel sequencing by synthesis using Illumina v.4 chemistry and paired-end 100-bp reads (Illumina).

RNA-seq data analysis

All RNA-seq data analyses were performed in R. v.3.6.1 and Bioconductor v.3.10. RNA-seq data were aligned to the reference human genome (build hg19) using STAR v.2.4.0e (24). Mapped reads were then used to quantify transcripts with HTSeq v.0.6.1p1 (25) and RefSeq GRCh38 v.82 gene annotation. The gene-expression data were normalized using the trimmed mean of *M*-value normalization as implemented in the edgeR R/Bioconductor package v.3.20.9 (26). Hierarchical clustering was performed using Euclidean distance for samples and Pearson correlation for genes and gene sets. Features were ordered using the optimal leaf-ordering algorithm as implemented in the R package cba v.0.2.19. The pathway/gene set expression was derived using the geometric mean expression of all genes in each set (Supplementary Table S1). Complex heat maps (27) were used to generate heat maps, and bar graphs and box plots were generated using the ggplot2 package (28).

Data availability

Novartis cannot provide access to patient-level data if there is a reasonable likelihood that individual patients could be reidentified. Phase I studies, by their nature, present a high risk of patient reidentification; therefore, RNA sequences could not be provided in this manuscript. However, the raw transcript number for each of the >19,200 genes examined in all patients, as well as in paired tumor and paired blood samples (taken pre- and post-dosing of NJH395), has been provided. This allows for the assessment of gene-expression changes with NJH395 across the transcriptome. This is contained in the Supplementary Data excel file.

Additional statistical analyses

Statistical analyses are described in the specific methods sections where applicable. The R software (open-source) was used to fit the Bayesian logistic regression model.

Results

Pharmaceutical properties and *in vitro* characterization

NJH395 is an ISAC composed of an immunoglobulin G1 (IgG1) antibody (MIW338) conjugated to an immunostimulatory TLR7 agonist linker payload. MIW338 is a humanized monoclonal antibody that binds HER2. It belongs to the IgG1/ κ isotype subclass and has a sequence similar to trastuzumab. Four cysteine residues were

introduced for site-specific conjugation with the TLR7 agonist linker payload via a maleimide ring (Fig. 1A) to provide increased stability (29). Sites were chosen to avoid disrupting binding to HER2 and Fc γ receptors. NJH395 bound specifically to human and cynomolgus monkey HER2 ($K_d = 0.33$ and $= 5.0$ nmol/L, respectively) and had a similar binding affinity to the anti-HER2–drug conjugate ado-trastuzumab emtansine (T-DM1; Supplementary Table S2). The similar affinity of NJH395 with well clinically established trastuzumab-based drugs, such as T-DM1, underscore the clinical relevance of NJH395.

In vitro incubation studies with HER2⁺ breast cancer cells HCC1954 confirmed that NJH395 and its unconjugated antibody component MIW338 bind to target cells, whereas isotype control molecules do not (Supplementary Fig. S1A). Internalization in HCC1954 cells was observed for NJH395, but not the isotype control ISAC NJH397 (Supplementary Fig. S1B). The linker payload in NJH395 is noncleavable to minimize the risk for payload release before binding to HER2⁺ tumor cells; *in vitro* incubation with N87 gastric cancer cells opened the maleimide ring, suggesting that upon internalization, NJH395 is lysosomally processed to yield active catabolites consisting mainly of the cysteine linker-payload adduct MIM697 and the minor catabolite YFB712 (Supplementary Fig. S2). *In vitro* assays were conducted with these catabolites to measure TLR activation. Because TLR activation facilitates transcription of genes controlled by NF- κ B promoters, human embryonic kidney cells (HEK293T) were engineered to express luciferase under the control of an NF- κ B promoter. The parental reporter line was further engineered to coexpress either TLR7 or TLR8 to assess the potency and selectivity of TLR7 agonism. Both NJH395 catabolites induced the NF- κ B reporter in TLR7-expressing HEK cells with maximum activation comparable to the control TLR7/8 agonist resiquimod (Fig. 1B). The main catabolite MIM697 exhibited no significant activity in the HEK-TLR8 reporter assay (Supplementary Fig. S3A), indicating TLR7 specificity. Because TLR7 induces transcription of proinflammatory cytokines (TNF α , IL6, IL8, and IL1 β), we used these cytokines as a measure of TLR7 activation in human PBMCs and coculture assays. MIM697 and YFB712 induced IFN α and IL6 secretion, suggesting robust TLR7 activation in human PBMCs (Supplementary Fig. S3B and S3C). Whole blood from healthy donors was incubated with NJH395 or its unconjugated anti-HER2 antibody component MIW338. qPCR revealed upregulation of genes for TNF α , IL6, and IL1 β for NJH395 but not for the unconjugated antibody, suggesting that the effects are driven by TLR7 agonism (Supplementary Fig. S3D). This is further supported by upregulation of *OAS1* and *OAS2*, which represent TLR7 downstream targets. The pharmacodynamic activity of NJH395 was further assessed in a human coculture assays of HER2-expressing HCC1954 tumor cells and THP-1 macrophages that lacked HER2 expression (Supplementary Fig. S3E). qPCR showed that *TLR7* was not expressed in HER2⁺ HCC1954 cancer cells (Supplementary Fig. S3F). After incubation with NJH395 or controls, supernatants were evaluated for IL1 β production. NJH395 induced significantly more IL1 β production when TLR7-expressing THP-1 macrophages and HER2-expressing HCC1954 cells were cocultured, compared with the isotype ISAC NJH397, whereas no induction of IL1 β was observed with the unconjugated anti-HER2 antibody (MIW338) or an unconjugated isotype control human IgG1 (Fig. 1C). However, NJH395 did not induce IL1 β after incubation with HCC1954 tumor cells alone, although it showed minimal activity in THP-1 cells alone (Supplementary Fig. S3G and S3H). The NJH395 catabolite MIM697 induced IL1 β secretion in TLR7-expressing macrophages and cocultures, but not in HCC1954 cells (Supplementary Fig. S3I). Furthermore, MIM697 induced NF- κ B phosphorylation in

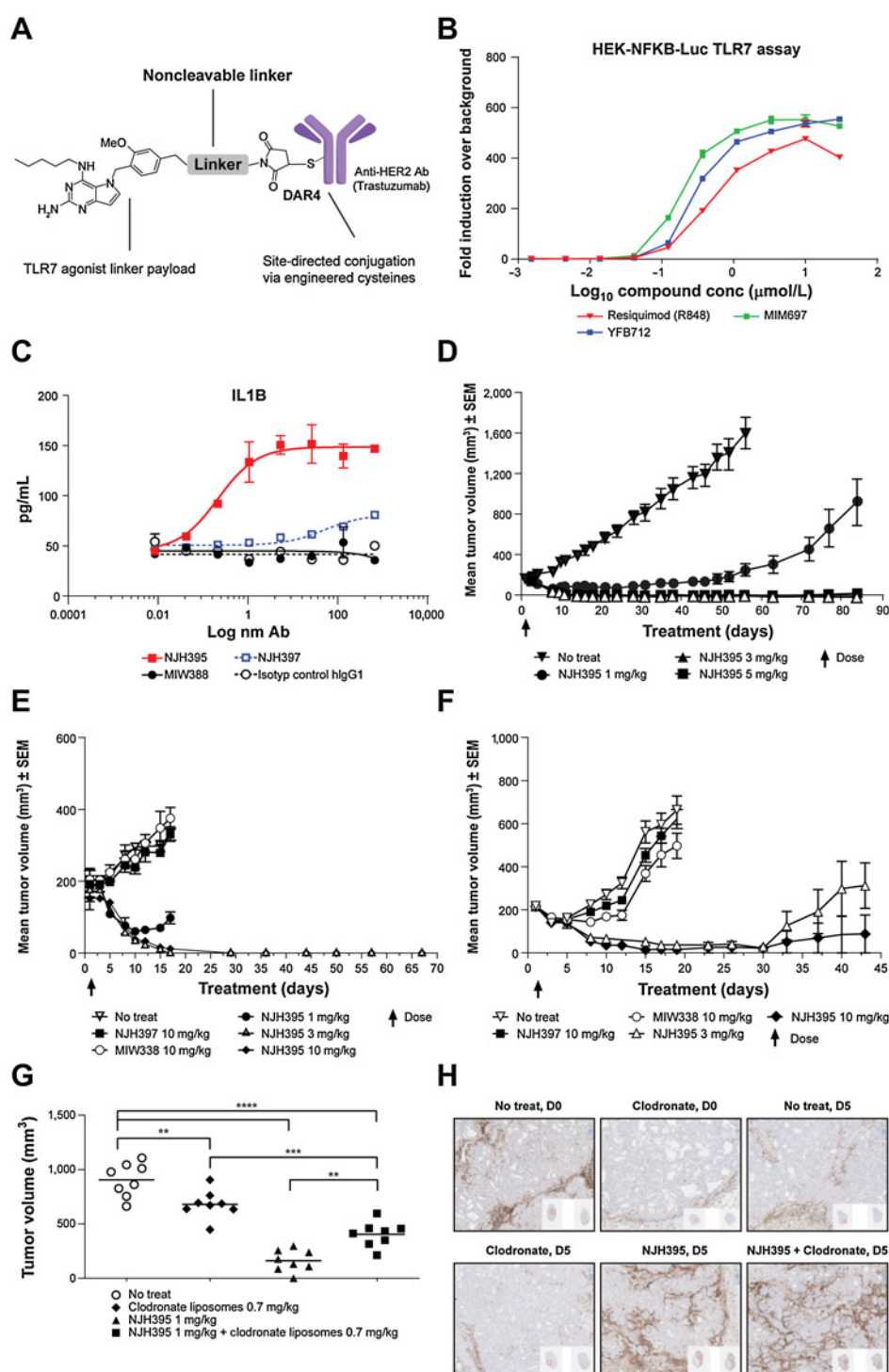


Figure 1.

Preclinical pharmacology data demonstrating NJH395 activity. **A**, Schematic of NJH395, illustrating the TLR7 agonist at a drug-to-antibody ratio (DAR) of approximately 4, noncleavable linker, and the HER2 antibody. **B**, Synthetic compounds, representing putative TLR7 agonist catabolites of NJH395 (MIM697, YFB712), were assessed for activity of luciferase downstream of an NF-κB reporter gene in a HEK293-TLR7 cell line. The TLR7/8 agonist, resiquimod, was used as a positive control (mean ± SD of triplicate measurements). **C**, Human THP-1 cells were treated with PMA to generate differentiated, adherent macrophage-like cells, and then cocultured with HER2⁺ tumor cells line HCC1954. Cocultures were treated with NJH395 or control molecules (unconjugated anti-HER2 antibody MIW388, isotype ISAC NJH397, and isotype control IgG) and evaluated for IL1β in the media following TLR7-driven cytokine secretion (mean ± SEM of duplicate measurements). **D–F**, *In vivo* efficacy of NJH395 or controls in the **(D)** gastric carcinoma N87 xenograft model, **(E)** the breast carcinoma HCC1954 model, and **(F)** the ovarian carcinoma SKOV-3 model using SCID beige mice following a single intravenous dose ($n = 8$ /group for all three models; mean ± SEM reported for each time point). Unconjugated anti-HER2 antibody MIW388 and isotype ISAC NJH397 were used as controls. (Continued on the following page.)

the THP-1–derived macrophages, suggesting TLR7 activation (Supplementary Fig. S3J and S3K). Together, these data suggest that NJH395 can more optimally activate myeloid cells in the presence of antigen-expressing cancer cells and that both cancer antigen targeting and TLR7 agonism contribute to this effect.

The minimal activity seen with the isotype NJH397 suggests that non-HER2-mediated mechanisms, such as FcγR function, may have a minor contribution to the ISAC activity. For further assessment, NJH395 was incubated with a THP1-Dual reporter cell line derived from human THP-1 monocytes. NJH395 showed activation of TLR signaling in the reporter assay, whereas no activity was observed with the unconjugated antibody (Supplementary Fig. S3L). Because THP-1 cells lacked HER2 expression, the data further support the hypothesis that TLR7 activation in myeloid antigen-presenting cells (APC) may involve additional FcγR-dependent mechanisms. To test this, we included a D256A-P329A (DAPA)–mutated version of NJH395. The DAPA version, which reduces FcγR binding (30), showed reduced potency in the reporter assay (Supplementary Fig. S3L). Further assessment in cocultures of THP-1 macrophages and HCC1954 tumor cells using the DAPA-mutated version and a variant S239D-A330L-I332E with increased FcγR binding (DLE; ref. 31) confirmed decreased activation of IL1β with DAPA, but no enhancement with the DLE variant, further supporting that FcγR interactions may contribute to, but do not drive, the ISAC activity (Supplementary Fig. S3M). The DAPA and DLE variants had the same binding affinity to HER2-expressing HCC1954 cells as the parental molecule NJH395 (Supplementary Fig. S1A).

***In vivo* antitumor activity in HER2-expressing xenograft models**

The *in vivo* activity of NJH395 was evaluated in HER2-expressing xenograft models, including subcutaneous human gastric carcinoma N87, orthotopic human breast carcinoma HCC1954, and subcutaneous human ovarian carcinoma SKOV-3 in SCID beige mice. NJH395 or controls were administered once intravenously. Compared with controls, NJH395 significantly inhibited tumor growth in all three HER2-expressing models (Fig. 1D–F). Complete responses were observed with NJH395 doses ≥ 3 mg/kg in N87 and HCC1954 tumors (Fig. 1D and E). No significant antitumor effects were observed for the unconjugated antibody MIW338 or the isotype control ISAC NJH397 (Fig. 1E–F), suggesting that both HER2 targeting and TLR7 agonism contributed to *in vivo* efficacy of NJH395. The efficacy in SCID beige mice supports the hypothesis that NJH395 might eliminate tumors, even in the absence of B-, T-, or NK-cell activity. To assess the myeloid cell contribution, phagocytic cells were depleted using clodronate-loaded liposomes, as described previously (32, 33). A single dose of NJH395 was administered intravenously together with clodronate liposomes administered intraperitoneally twice a week in N87 tumor-bearing SCID beige mice and compared with control animals (untreated or treated with single agent). Mice treated with NJH395 alone had significantly smaller tumors on day 28 compared with untreated mice or mice treated with clodronate liposomes and NJH395

(Fig. 1G), suggesting a role for macrophages in NJH395 efficacy. Tumor infiltration by Iba-1⁺ macrophages was reduced following one or two doses of clodronate liposomes alone but increased following NJH395 treatment, even in combination with clodronate liposomes (Fig. 1H). The increase of macrophage infiltration even with clodronate pretreatment could explain the partial reduction noted with combination treatment. Iba-1 expression after treatment with NJH395 or controls was also investigated in N87 tumors growing in female Foxn1 nude mice. IHC showed that NJH395-treated tumors had higher infiltration by Iba-1⁺ macrophages compared with no treatment or treatment with the unconjugated antibody MIW338 or the isotype control ISAC NJH397 (Supplementary Fig. S3N), suggesting that HER2-mediated targeting of the TLR7 agonist in tumor cells was important for efficient recruitment and activation of myeloid cells in the TME.

NJH395 shows preclinical potential for cytokine release and immunogenicity

NJH395 was assessed in single- and repeat-dose toxicology studies in cynomolgus monkeys (see Materials and Methods). Systemic cytokines were measured using a Luminex-based multiplex assay or Meso Scale Diagnostics (MSD) custom oligoplex assay. Cytokines were increased in NJH395-treated monkeys. Moreover, IL6 and IFNγ-inducible protein 10 (IP-10) increased at 96 hours after the first dose but were mostly prominent 6 to 12 hours after the second dose with dose-related patterns (Supplementary Fig. S4A and S4B). We observed high cytokine induction by NJH395, NJH397, and MIM697, consistent with TLR7 agonism (high IL1RA, IL6) in human whole blood (Supplementary Fig. S4C). The isotype control NJH397 induced similar cytokine responses as NJH395. MIW338 showed small increases compared with controls at 6 and/or 24 hours of incubation in some cytokines, suggesting a limited role of specific HER2 antigen targeting in systemic cytokine release, possibly involving Fc receptors.

Studies in monkeys revealed ADAs against NJH395 in all animals. ADAs were detected 96 hours after administration and were more prominent after subsequent doses (Supplementary Fig. S5A). ADAs were detected for both NJH395 and the control isotype NJH397. ADA binding to the antibody and the TLR7 agonist was investigated using surface plasmon resonance technology. A prominent signal was observed with an immobilized molecule structurally identical to the NJH395 catabolite MIM697, suggesting that a significant ADA portion was directed against the payload.

Preclinical PK evaluation in monkeys showed that the NJH395 total ISAC (anti-HER2 antibody conjugated with at least one TLR7 agonist) and total antibody (anti-HER2 antibody with or without TLR7 agonist) PK profiles were comparable after the first dose (Supplementary Fig. S5B). Ninety-six hours after the first dose, an increase in clearance of total antibody and total ISAC exposure was evident. Because this occurred concurrently with ADA detection, drug exposure might have been affected by ADAs. Consistent with ADA impact, rapid NJH395 clearance was observed after the second and third administration (Supplementary Fig. S5). PKs of the small-molecule payload catabolite

(Continued.) **G**, Tumor size on day 28 following a single intravenous dose of NJH395 or PBS control (“no treat”) with or without phagocytic cell depletion by clodronate liposome treatment in the gastric carcinoma N87 model using SCID beige mice ($n = 8/\text{group}$). Clodronate liposomes were dosed on days –4, 3, and 7 relative to a single intravenous dose of 1 mg/kg NJH395 dosing. Significance was assessed between groups (**, $P < 0.01$; ***, $P < 0.001$; ****, $P < 0.0001$). Multiple group vs. group analysis: Significant differences were measured using one-way ANOVA, post-test Sidak multiple comparisons test (**, $P < 0.01$; ***, $P < 0.001$; ****, $P < 0.0001$). **H**, Observation of macrophage depletion following treatment with clodronate liposomes using Iba-1 staining by IHC. Tumors were collected on day 0 and on day 5 with or without a single intravenous injection of 1 mg/kg NJH395. Exemplary images are shown based on the assessment of $n = 2$ tumors per group per time point. Magnification is 40×.

MIM697 showed accumulation upon repeat dosing, which correlated with ADA increases, suggesting enhanced catabolism of NJH395 owing to ADAs and/or decreased payload clearance due to its binding to ADAs (Supplementary Fig. S5A).

Anatomic pathology findings in monkeys were primarily due to TLR7-related cytokine release, resulting in acute-phase response with correlating mononuclear cell infiltration in the spleen and bone marrow. Neuroinflammation was noted in one animal administered a 30 mg/kg single intravenous bolus. Inflammation was most pronounced in the corona radiata but also affected blood vessels of the gray matter (cerebral cortex) and meninges. Neuroinflammation was characterized by mixed inflammatory cells (predominantly monocytes/macrophages, less CD3⁺ T cells, neutrophils, and CD19⁺ B cells) in the vessel walls, meninges, and parenchyma surrounding the vessels (Supplementary Fig. S6A). *In situ* hybridization showed increased vascular/perivascular expression of IL6 and IL1 β mRNA (Supplementary Fig. S6B).

SAD phase I clinical trial: patient characteristics, safety, and efficacy

The phase I study was designed to include a SAD part (NCT03696771). Eighteen patients with non-breast HER2⁺ advanced malignancies were enrolled in SAD between December 2018 and April 2020, with last visit in October 2020. All patients were evaluable for safety after receiving one NJH395 dose. Eleven patients (61.1%) had metastatic HER2⁺ colorectal cancer. Patients

were enrolled in five dose cohorts (0.1–1.6 mg/kg; **Table 1**). Seventeen patients reported 132 treatment-related adverse events (AE; Supplementary Fig. S7). The most common AEs within 30 days of treatment suspected to be related to NJH395 were cytokine release syndrome (CRS; $n = 10$, 55.6%), pyrexia ($n = 8$, 44.4%), nausea ($n = 8$, 44.4%), vomiting ($n = 6$, 33.3%), headache ($n = 6$, 33.3%), chills ($n = 5$, 27.8%), increased aspartate aminotransferase (AST; $n = 4$, 22.2%), and increased alanine aminotransferase (ALT; $n = 4$, 22.2%; **Fig. 2A**). Most CRS events started 12 to 24 hours after NJH395 treatment. All CRS events were grade 1 or 2 and resolved after treatment per institutional guidelines, including treatment with corticosteroids ($n = 5$) and tocilizumab in a patient with grade 2 CRS. Five DLTs were reported in 3 patients: (i) grade 3 increased AST in a 46-year-old male patient with metastatic esophageal adenocarcinoma treated with NJH395 at 0.2 mg/kg; (ii) grade 3 meningitis with symptoms of headache, photophobia, and neck stiffness in a 45-year-old female patient with metastatic colon adenocarcinoma treated with 1.6 mg/kg; and (iii) grade 3 meningitis (aseptic) and grade 3 increased ALT and AST in a 72-year-old male patient with metastatic rectum adenocarcinoma treated with 1.6 mg/kg. All DLTs resolved after treatment per institutional guidelines.

Seventeen patients were evaluable for efficacy at the end-of-treatment (EOT) visit on day 21 after NJH395 dosing. One patient treated with 1.6 mg/kg had no postbaseline tumor assessment owing to treatment-site restrictions related to the coronavirus disease 2019

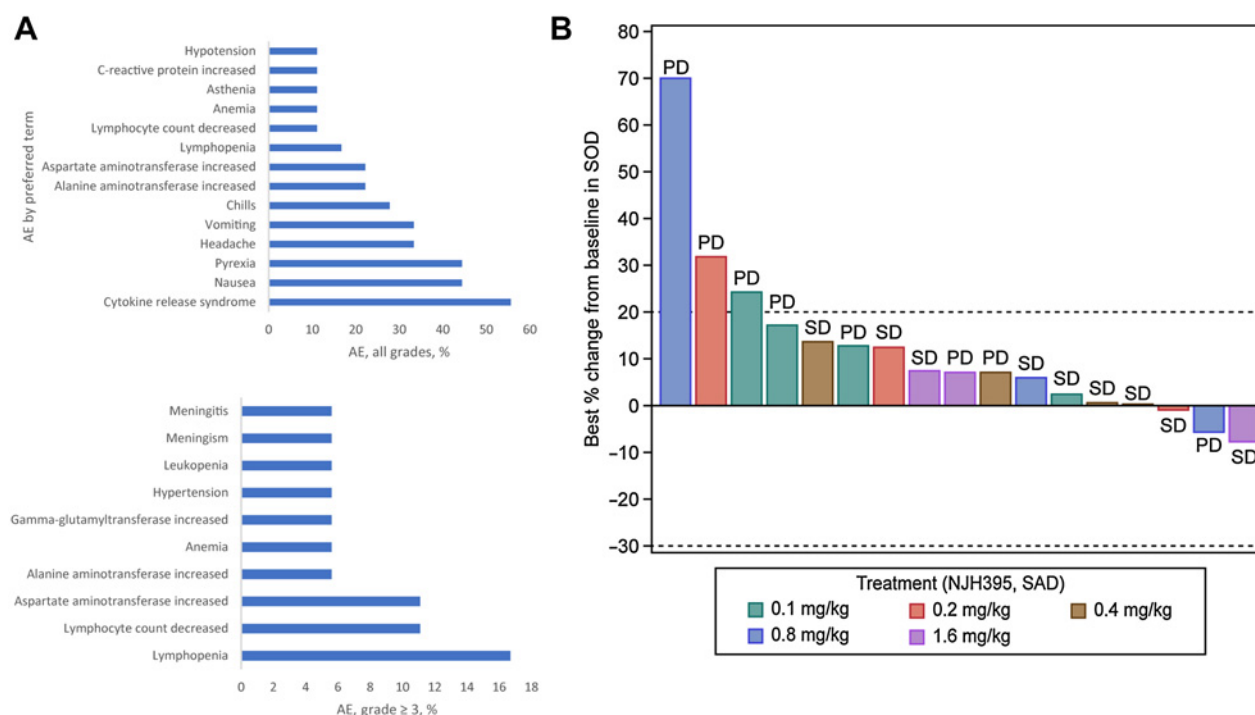


Figure 2.

Clinical safety and efficacy of single-dose NJH395 in patients with HER2⁺ advanced malignancies. **A**, Summary of treatment-related adverse events (TRAE) reported in ≥ 2 patients receiving NJH395 and summary of all grade ≥ 3 TRAEs per Common Terminology Criteria for Adverse Events (CTCAE) version 5.0. **B**, Best percentage change from baseline in SOD (sum of diameters) of target lesions and best overall response using RECIST version 1.1 at the end of the treatment visit, 21 days after single dosing. $N = 17$; one patient treated with NJH395 at a dose of 1.6 mg/kg had no post-baseline tumor assessment due to COVID-19. SD, stable disease; PD, progressive disease; SAD, single ascending dose. **Figure 2** was produced in R 3.6.3. The dotted line represents the reference line: 20% increase for PD criteria and 30% decrease for PR criteria.

(COVID-19) pandemic. **Figure 2B** presents the best percentage change from baseline. No complete or partial responses per RECIST version 1.1 were observed. Nine patients had stable disease at EOT. Follow-up efficacy beyond EOT was collected for 5 patients showing progressive disease in all cases (Supplementary Fig. S8).

PKs and immunogenicity

The total ISAC PK profile showed a greater than dose-proportional increase in exposure (AUC_{inf} and C_{max}) with increasing dose of NJH395, suggesting more rapid target-mediated drug disposition at lower dose levels (**Fig. 3A**). Total ISAC clearance was 34.6 to 89.1 mL/hour, with clearance decreasing with increasing dose levels. Low levels

of the payload catabolite MIM697 were detected in plasma only for NJH395 doses of 0.8 and 1.6 mg/kg (**Fig. 3B**). Supplementary Table S3 summarizes PK parameters.

Immunogenicity assessment revealed that all 18 patients developed ADAs to NJH395. Preexisting ADAs were detected in 1 patient. In all other patients, ADA detection started at the first timepoint evaluated post-dose (96 hours), with high titer increases for doses ≥ 0.2 mg/kg (**Fig. 3C**). Neutralizing antibodies against NJH395 were detected in 8 of 18 patients (44.4%) using a competitive ligand-binding assay with recombinant HER2. Neutralizing antibodies against trastuzumab were detected in 3 of 18 patients (16.7%; 2/18 transient, 1/18 persistent; Supplementary Fig. S9).

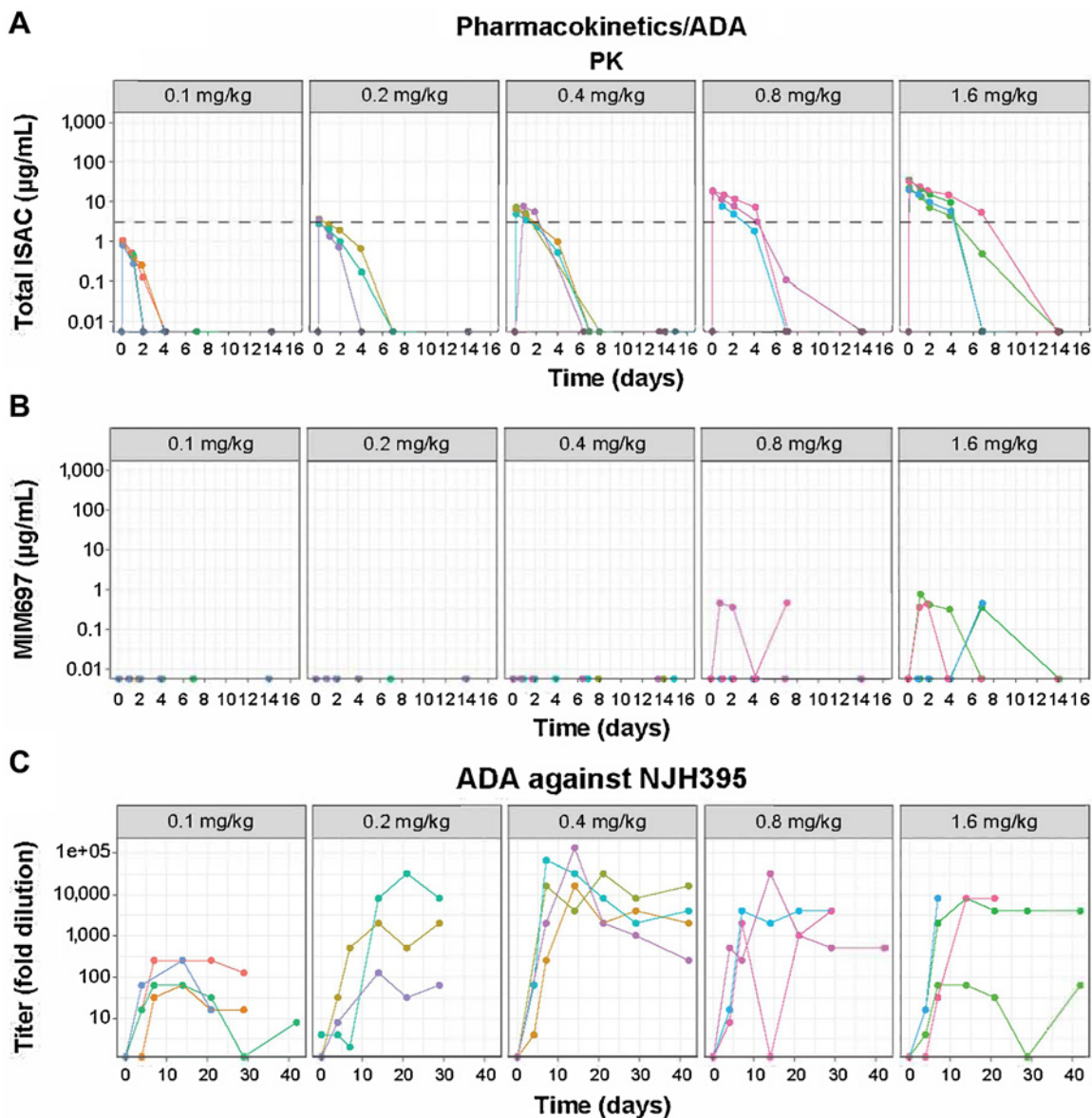


Figure 3.

PKs and immunogenicity of NJH395 in patients with HER2⁺ advanced malignancies. **A**, PKs of the total ISAC in plasma of treated patients at the indicated doses. **B**, Pharmacokinetics of the TLR7 agonist catabolite MIM697 in plasma. **C**, ADA (antidrug antibody) titers against NJH395 in patients treated with NJH395 in the SAD (single ascending dose) clinical trial; $N = 18$.

Clinical PKs and immunogenicity aligned with preclinical data. Toxicities and ADAs may have restricted the use of higher doses in humans that were predicted for maximal therapeutic benefit. Similar to preclinical results, a significant portion of the anti-NJH395 ADA response in humans was directed against the TLR7 agonist payload. PKs in patients showed an increase in clearance after 96 hours, coinciding with ADA detection, suggesting that ADAs might affect NJH395 exposure.

Biomarker and pharmacodynamic data

Biomarker analyses were performed in paired tumor biopsies collected at baseline and on-treatment (C1D5 $\pm$ 3 days). Paired tumor biopsies were collected from 16 of 18 patients enrolled in SAD (Supplementary Table S4). All 16 paired tumor samples were evaluable for IHC. Fourteen paired tumor biopsies and 12 paired blood samples were evaluable for RNA-seq. IHC was performed for CD8⁺ T cells, CD68⁺ macrophages, HER2, programmed cell death-

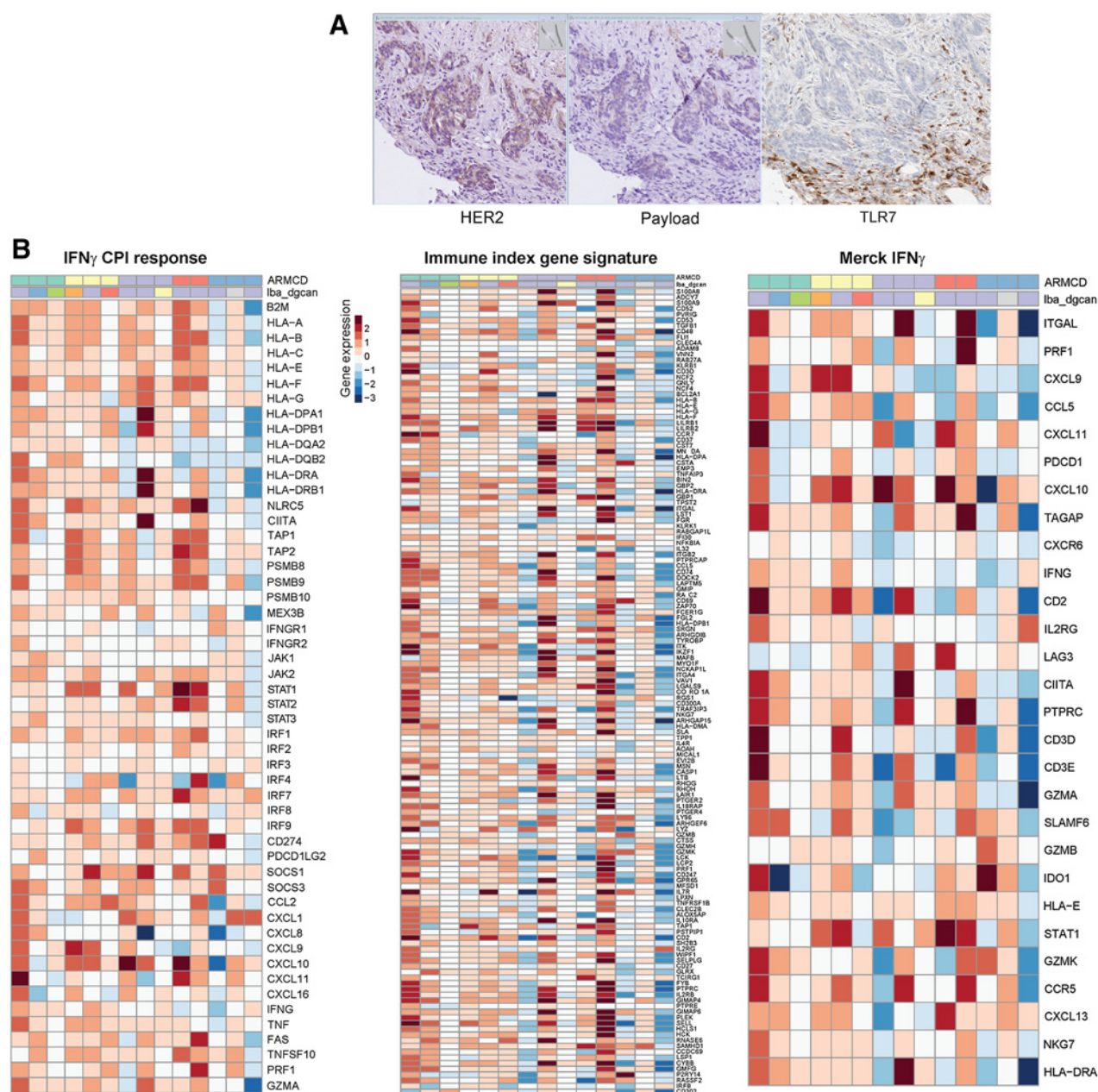


Figure 4.

Biomarker and pharmacodynamic analysis. **A**, IHC of HER2 (left), NJH395 payload (middle), and TLR7 (right) in an on-treatment biopsy from a patient treated with NJH395 at 1.6 mg/kg. Magnification is 40 $\times$. **B**, RNA-seq was performed on paired tumor biopsies collected at baseline and on-treatment with NJH395. Shown are multiple immune gene signatures (39–41). Treatment arm and cancer type are indicated. Log₂ fold change in on-treatment vs. baseline tumor samples is shown. On-treatment biopsies were collected on day 5 ($\pm$ 3 days) after a single treatment with NJH395. (Continued on the following page.)

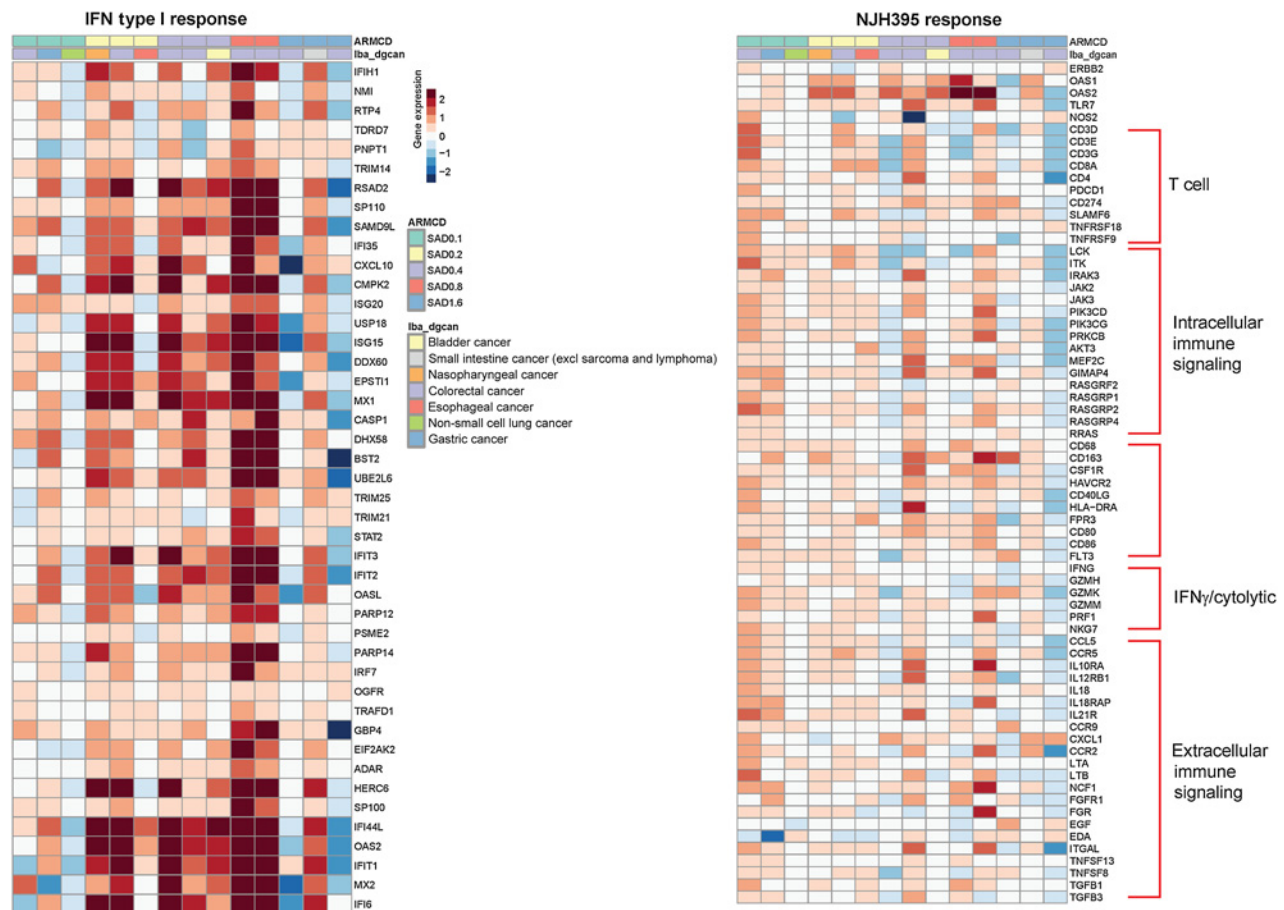


Figure 4.
(Continued.)

ligand 1 (PD-L1), TLR7, and the NJH395 payload. Consistent with preclinical findings and with the expected impact of TLR7 engagement in myeloid cells, we observed an increase in CD68⁺ cells following treatment with NJH395 in tumors from patients across different dose levels (Supplementary Fig. S10). The number of CD8⁺ T cells in tumors increased after NJH395 treatment in a subset of patients (Supplementary Fig. S11). In both cases, no dose-response relationship in the magnitude of cellular response was observed. IHC demonstrated NJH395 payload detection in six on-treatment tumor samples. Staining was cytoplasmic and correlated with regions of HER2 tumor staining (Fig. 4A). Payload was detected in tumors with higher HER2 scores after treatment at doses ≥ 0.2 mg/kg (Supplementary Fig. S12). Investigation of TLR7 expression by IHC revealed an increase in on-treatment tumor biopsies in 8 of 13 patients. TLR7 was predominantly detected in macrophages in the TME (Fig. 4A), consistent with studies suggesting that type I IFN activation induces TLR7 expression in macrophages (34, 35). Increased TLR7 expression was detected in 5 of 6 tumors positive for NJH395 payload (Supplementary Fig. S13). Together, these data link the detection of a systemically delivered TLR7 agonist in tumors to the expression of a specific targeting antigen (HER2) and a receptive cell population (myeloid cells), providing, for the first time, clinical evidence that supports the ISAC overarching concept.

RNA-seq in paired tumor biopsies showed induction of multiple immune gene signatures after NJH395 treatment, including signatures for type I IFN response, IFN γ response, immune index, and signatures encompassing broad lists of immune-related genes, including genes related to APC activation, cytokines/chemokines and their receptors, antigen presentation, intracellular, and extracellular immune signaling, NK-cell activity, and T-cell markers (Fig. 4B). Upregulation of type I IFN response genes reflecting downstream TLR7 cascades was observed in 9 of 14 tumors, indicating target engagement. RNA response did not exhibit dose dependency. On-treatment gene-expression changes in tumors were compared with changes in matched blood samples in 12 patients. The comparison of immune signatures capturing changes that occur following the triggering of type I IFN response revealed greater upregulation in paired tumor biopsies compared with paired blood samples in a subset of patients (Supplementary Fig. S14A and S14C), consistent with the hypothesis of preferential TME modulation. The type I IFN response signature was also upregulated in 6 of 12 blood samples (Supplementary Fig. S14D). Upregulation in blood could either reflect the engagement of circulating mononuclear cells through Fc γ R-dependent targeting or a secondary cascade of immune activation triggered by reactions in tumors. Figure 5 presents IHC and RNA-seq findings for each patient to evaluate the interrelation of different components of NJH395-mediated immune activation in tumors. Because preclinical

studies revealed increases in CD68⁺ cells in tumors following ISAC treatment, data were grouped by tumor changes in CD68⁺ cells (9 patients with CD68⁺ cell increase trend vs. 5 patients with decrease trend). The data indicated that the CD68⁺ cell increase was interrelated in several patients with TLR7 induction in both IHC and RNA-seq. Moreover, these changes were interrelated with signs of target engagement, such as upregulation of type I IFN response, OAS1, and OAS2, as well as with signs of immune modulation in the TME, such as upregulation of IFN γ , immune index, and an NJH395 response signature (Fig. 5). PD-L1 increases in a subset of patients were also consistent with TME modulation. All five tumor samples with the highest NJH395 payload detection exhibited evidence of immune modulation/activation (Fig. 5). Together, these findings support the hypothesis that immune components are activated in HER2⁺ tumors owing to preferential ISAC-mediated TLR7 engagement.

Although some induction of type I IFN and IFN γ response signatures was observed in blood, this was associated with the upregulation of immune responses in tumors (Fig. 5). Whether this association was related to a secondary cascade of immune activation in blood triggered by reactions in the TME or was an effect of circulating mononuclear cell engagement through Fc γ R-dependent targeting remains unclear and warrants further investigation. Our analysis did not show an association between immune activation in tumors and CRS; approximately half of the patients who developed CRS did not show immune responses in their tumors, and immune activation in tumors was also observed in patients without CRS, suggesting that immune effects in tumors were not secondary to systemic inflammatory responses. Likewise, the use of corticosteroids before on-treatment biopsy did not correlate with lack of immune activation in tumors. However, it may have affected immune signatures in blood, as 6 of 7 patients who received steroids did not exhibit upregulation of relevant genes in their blood. The raw transcript counts for the 19,239 genes examined in the paired tumor and blood samples are provided in the Supplementary Data excel file.

Systemic cytokine release was measured in plasma at baseline, at 6 hours, on day 5, and day 15 after NJH395 treatment. Exploratory analysis demonstrated increases in inflammatory cytokines (IP-10, IL6, IL8, MIP1 β , TNF α , and IFN γ) in plasma following ISAC treatment (Supplementary Fig. S15). Cytokine increases were observed across dose levels with limited dose–response relationship. Overall, greater IP-10 was observed in patients treated with higher NJH395 doses, whereas a clearer dose–response relationship was observed for the macrophage inflammatory protein MIP1 β (Supplementary Fig. S15). Increases in IL8 (a neutrophil chemotactic factor) associated with transient increases in neutrophil counts in peripheral blood (Supplementary Fig. S16).

Discussion

This study presents the translational and clinical assessment of a novel ISAC containing a HER2-targeting antibody attached to a TLR7 agonist payload through a noncleavable linker. This targeted strategy aimed to preferentially deliver a TLR7 agonist into tumors, while lowering systemic exposure and minimizing immune-related toxicities. This is to our knowledge the first clinical trial to test ISACs in humans. Our comprehensive evaluation, including multimodal (IHC, RNA-seq) biomarker assessment of pharmacodynamics in patients revealed for the first time that ISACs could preferentially deliver immune agonists in tumors and promote targeted immune activation in the TME. This study also demonstrated that both tumor antigen

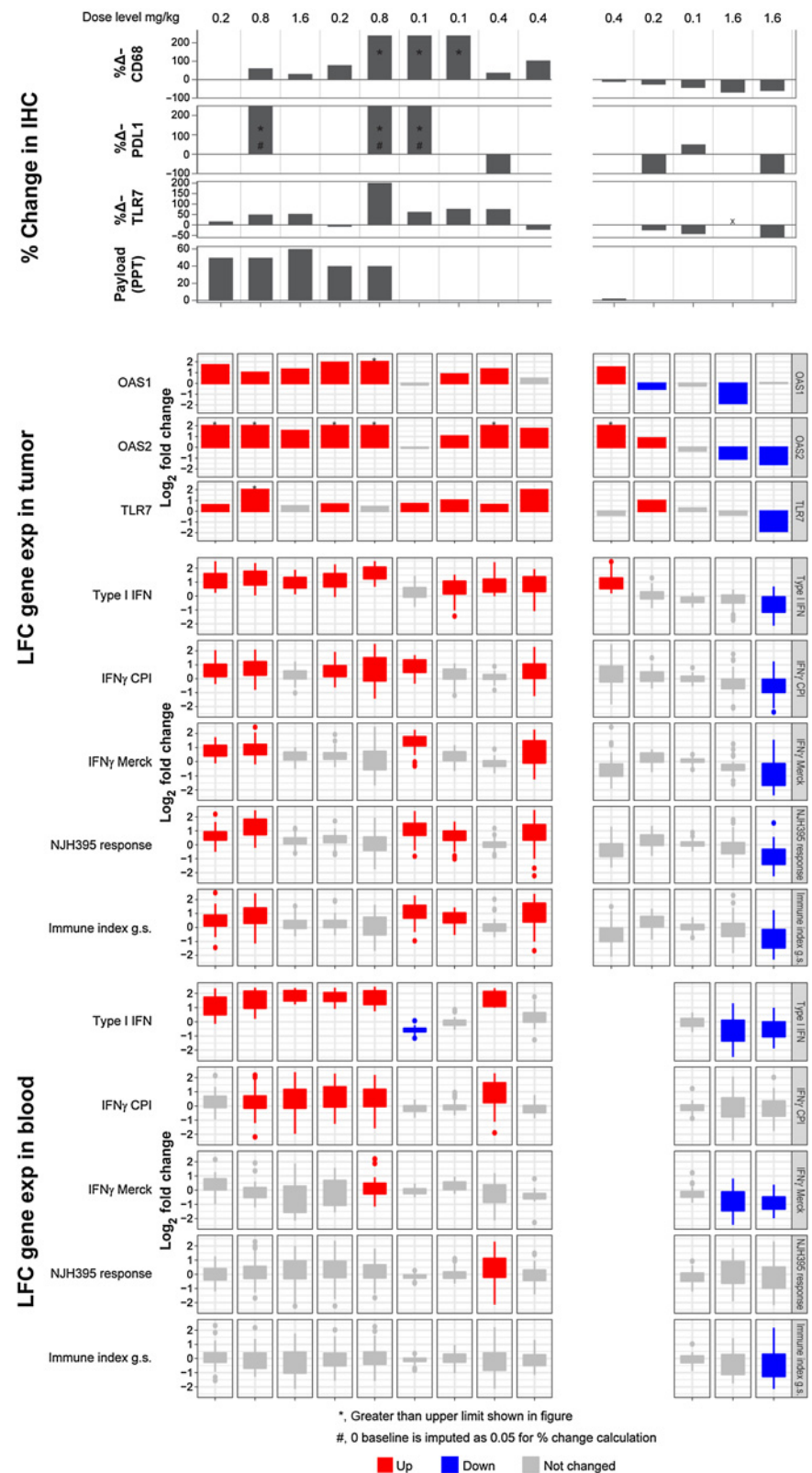
targeting and payload-mediated agonism contributed to immunostimulatory effects, which might facilitate further development of novel ISACs with improved targeting properties and optimized linker payloads. Paired tumor biopsies demonstrated payload detection in HER2-expressing tumor cells and revealed a TLR7 increase mainly in macrophages in the TME. Clinical biomarkers demonstrated an upregulation of type I IFN in tumors, suggesting target engagement, as type I IFN response reflects downstream TLR7 cascades (36, 37). Type I IFN activation was interrelated with intratumoral immune modulation. CD68⁺ cells showed an increasing trend in most tumors after treatment with NJH395, consistent with NF- κ B-mediated proliferative signaling following TLR7 engagement in myeloid cells (38). Increases in CD8⁺ T cells in a subset of tumors allow the hypothesis that preexisting T cells might be boosted through the innate activation of myeloid cells in the TME. However, changes in CD8⁺ T cells in our study were unlikely to reflect *de novo*-generated responses, as the on-treatment biopsy timeframe was likely inadequate for migration of activated DCs to regional lymph nodes where T-cell priming may occur, followed by the trafficking of stimulated T cells back to the tumor. Tumor immune modulation is further supported by PD-L1 increases in some cases, as well as by upregulation of established gene signatures encompassing broad immune-related genes. These included IFN γ response signatures associated with antigen presentation, chemoattractants, cytotoxic effectors, and signaling (39) and a signature (IFN γ -Merck) encompassing 28 genes related to cytolytic activity (e.g., granzyme A/B/K, PRF1), cytokines/chemokines for initiation of inflammation (CXCR6, CXCL9, CCL5, CCR5), T-cell markers (CD3D, CD3E, CD2, IL2RG), NK-cell activity (NKG7, HLA-E), antigen presentation (CIITA, HLA-DRA), and additional immunomodulatory factors (LAG3, IDO1, SLAMF6; ref. 40). Together, these data provide the first clinical evidence linking the detection of an immune agonist payload delivered to antigen-expressing tumors, increasing a receptive cell population, and enhancing the immune response after systemic ISAC administration. Providing such critical scientific and translational insights into the biology of a novel class of drugs is of high significance, as unraveling mechanisms of ISACs could facilitate the development of optimized immune therapeutics.

In a recent study, the immune index gene signature (41) was increased in patients with breast cancer who received a single trastuzumab dose and attained pathologic complete response (42). Notably, the signature described in Grasso and colleagues (39) has been clinically validated to have predictive value for clinical outcome following anti-PD-1 treatment in metastatic melanoma. A modified signature including 18 genes from that list demonstrated a positive association with objective responses and progression-free survival after treatment with pembrolizumab in other cancer indications (40). Various studies confirmed that response to anti-PD-1 treatment likely associates with a preexisting IFN γ transcriptome signature (39, 43). Preliminary data from intratumoral application studies suggest that TLR agonists might enhance anti-PD-1 therapy or reverse anti-PD-1 resistance (44–46). Because NJH395 promotes the induction of IFN γ genes, our findings allow the hypothesis that TLR7-based ISACs could sensitize tumors to ICI, opening the path for novel immunotherapy combinations in the future.

Our study reveals critical challenges that may broadly apply to ISACs and should be critically considered during their development. The biomarker-based evidence of proinflammatory TME modulation was not associated in our trial with clinical responses. This may be because only one dose of NJH395 was administered to patients, which

Figure 5.

Summary of IHC and RNA-seq in on-treatment vs. baseline tumor biopsies by patient shows the interrelation of pharmacodynamic markers. Data were grouped by tumor changes in CD68⁺ cells for patients with a CD68⁺ increase trend ($n = 9$, left) vs. patients with no increase/decrease in CD68⁺ cells in the tumor microenvironment ($n = 5$, right) after a single treatment with NJH395. On-treatment biopsies were collected on day 5 (± 3 days) after single dosing of NJH395. LFC, Log₂ fold change; PPT, percent positive tumor. **Figure 5** for IHC data was produced in R 3.6.3. Box of the RNA box plot displays 25th to 75th percentiles.



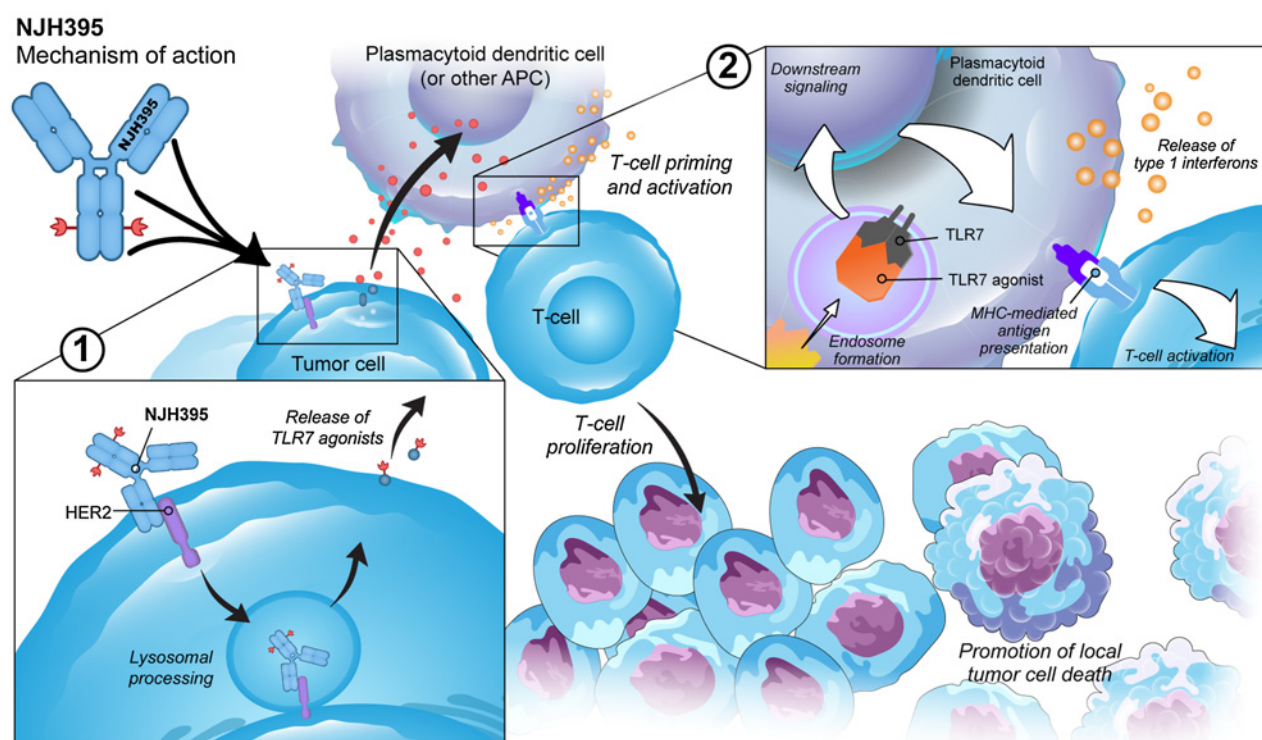


Figure 6.

Proposed mechanism of action of the anti-HER2-TLR7 ISAC NJH395. NJH395 is an ISAC comprising an anti-HER2 monoclonal antibody (a cysteine-engineered variant of trastuzumab, MIW338) covalently linked to a TLR7 agonist. NJH395 binds to HER2 on the tumor cell surface, followed by payload uptake in APCs through either payload catabolite release in the TME or phagocytosis (1). The activation of TLR7 in APCs induces a type I IFN response that results in preferential immunomodulation of the TME (2), characterized by myeloid cell proliferation and T-cell activation, promoting tumor-specific immunologic memory and antitumor response.

might not have induced tumor inflammation of sufficient magnitude and duration. Notably, reports from repeat dosing of other HER2-targeting ISACs provide evidence of modest antitumor activity (47). Other factors that may have contributed to the lack of correlation may include the small number of patients in the SAD part of the study, as well as the fact that the majority of patients (13 of 18) had tumor types such as colorectal cancer ($n = 11$), pancreatic cancer, and small intestine carcinoma, where the effectiveness of immunotherapies has not been clearly established beyond certain patient subsets such as dMMR-MSI-H colorectal cancer, which account for ~15% of colorectal cancer cases. Preclinical and clinical studies revealed ADA formation following treatment with NJH395, which is associated with decreases in drug exposure. Although the uniform induction of ADAs is not desirable in a therapeutic approach, it provides further proof-of-mechanism for ISACs, as it highlights ways in which coupling of innate immune activators to defined proteins could serve as a potent vaccination strategy. The unfavorable benefit-to-risk ratio owing to ADA-mediated drug exposure decrease underscores the need for detailed ADA assessment and development of ISACs with decreased antidrug immunogenicity.

The safety profile of NJH395 might inform the development of other novel ISACs. Consistent with the mechanism of action of TLR agonists, our study identified CRS as a key safety risk. Although CRS events occurred in patients across all dose levels without dose-toxicity relationship, all were low-grade and resolved after treatment, indicating that CRS after ISAC treatment is manageable. Moreover,

DLTs included two cases of meningeal inflammation. Neuroinflammation following CNS infections is associated with PAMPs that stimulate TLRs, and TLR7 can elicit antiviral effects and induce neuroinflammatory responses (48–50). Previous preclinical studies report pronounced inflammatory responses in the brain 12 hours after injecting TLR7 agonists into the lateral ventricles of mice, with mRNA upregulation for cytokines/chemokines (TNF, MCP-1, and IP-10) and neutrophil infiltration. The same studies report that cells in the choroid plexus and brain capillary endothelial cells are among the cell types that respond to TLR7 stimulation (48). Interestingly, the effects of TLR7 are dose dependent, and mRNA expression returns to basal levels after 48 hours (49). Both patients with meningeal inflammation in our study received NJH395 at the highest dose level, whereas some patients treated at lower doses reported nonspecific symptoms, including headaches, suggesting dose dependency of neuroinflammatory effects. Corroborating previous preclinical findings, both patients exhibited increases in polymorphonuclear cells in the cerebrospinal fluid. Clinical symptoms started 12 to 24 hours posttreatment and resolved after 2 to 5 days, seemingly in line with the onset and transient duration of the reported preclinical neuroinflammation. Inflammatory brain changes were observed in a monkey treated with high-dose NJH395; the brain tissue analysis revealed mixed inflammatory cells in the vessel walls. Hence, a hypothesis that warrants further investigation is that high-dose TLR7-based ISACs might stimulate brain endothelial cells, leading to cytokine/

chemokine release, which induce neuroinflammation and blood-brain barrier disruption. Although dose dependency is suggested for severe AEs (i.e., meningism/meningitis), the tumor biomarkers did not demonstrate a dose-response relationship. It is possible that tumor immune responses to ISACs could be organized more like a binary switch, activated after a threshold is reached. The threshold for immune activation in tumors could be reached at lower doses than those required to induce neuroinflammation. This suggests that dose escalation to maximum-tolerated dose levels might not be required for TLR7-based ISACs to exert their activity in the TME, providing a mitigation strategy for safety risks in future ISAC development.

Our findings align with preclinical studies reporting that molecules comprising a TLR7/8 dual agonist conjugated to HER2-targeted antibodies trigger localized immune responses, resulting in tumor regression and immunologic memory (33). Consistent with our results, these studies demonstrate that tumor antigen recognition and TLR-mediated myeloid activation contribute to antitumor activity and suggest a role for FcγR-mediated phagocytosis. Building on promising preclinical findings, we provide the first clinical evidence for proof-of-mechanism of ISACs. Mechanistically, the anti-HER2 antibody delivered the TLR7 agonist in HER2⁺ tumor cells, followed by payload uptake in APCs, either through payload catabolite release in the TME or phagocytosis. Subsequent activation of TLR7 induced a type I IFN response that resulted in preferential immunomodulation of the TME, characterized by myeloid cell proliferation and T-cell activation (Fig. 6). However, our study also revealed significant challenges in the clinical application of ISACs, which should be overcome for the successful ISAC application in the clinical setting. Besides improvements that mitigate the risks for ADAs and inflammatory responses, identification, and selection of patients likely to benefit from ISAC-mediated tumor immunomodulation is needed. Although our data provide critical insights into the ISACs' mechanism of action, they do not allow addressing the question of whether TLR7 agonism can elicit *de novo* immune responses. Making the TME more immune-permissive with an ISAC could potentially be advantageous for patients with T-cell-inflamed tumors or circulating antitumor T cells. Hence, further investigation to unravel cancer immunobiological features predictive of responsiveness to ISAC-based immunotherapies is warranted to develop clinically efficacious ISACs.

Authors' Disclosures

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F. Janku: Resources, investigation, writing—original draft, writing—review and editing. **S. Han:** Resources, investigation, writing—original draft, writing—review and editing. **T. Doi:** Conceptualization, data curation, formal analysis, investigation, methodology, writing—original draft, writing—review and editing. **A. Amatu:** Resources, investigation, writing—original draft, writing—review and editing. **J.A. Ajani:** Resources, investigation, writing—original draft, writing—review and editing. **Y. Kuboki:** Investigation, writing—original draft, writing—review and editing. **A. Cortez:** Conceptualization, supervision, validation, investigation, visualization, methodology, writing—original draft, writing—review and editing. **S.E. Cellitti:** Data curation, investigation, methodology, writing—original draft, project administration, writing—review and editing. **P.C. Mahling:** Data curation, software, formal analysis, visualization, methodology, writing—original draft, writing—review and editing. **K. Subramanian:** Conceptualization, data curation, formal analysis, investigation, visualization, methodology, writing—original draft, writing—review and editing. **H.A. Schoenfeld:** Methodology, writing—original draft, writing—review and editing. **S.M. Choi:** Data curation, formal analysis, visualization, methodology, writing—original draft, writing—review and editing. **L.A. Iaconis:** Data curation, formal analysis, visualization, methodology, writing—original draft, writing—review and editing. **L. Lee:** Data curation, software, formal analysis, investigation, visualization, writing—original draft, writing—review and editing. **M.R. Pelletier:** Conceptualization, data curation, formal analysis, investigation, methodology, writing—original draft, writing—review and editing. **G. Dranoff:** Conceptualization, formal analysis, supervision, writing—original draft, writing—review and editing. **V. Askoxylakis:** Conceptualization, data curation, formal analysis, supervision, validation, investigation, visualization, methodology, writing—original draft, project administration, writing—review and editing. **S. Siena:** Resources, investigation, writing—original draft, writing—review and editing.

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